

# TPS65131-Q1 Positive- and Negative-Output DC-DC Converter

## 1 Features

- Qualified for automotive applications
- AEC-Q100 test guidance with the following results:
  - Device temperature grade 2:  $-40^{\circ}\text{C}$  to  $105^{\circ}\text{C}$  ambient operating temperature range
  - Electrical characteristics tested over  $-40^{\circ}\text{C}$  to  $125^{\circ}\text{C}$  junction temperature range
  - Device HBM ESD classification level H1C
  - Device CDM ESD classification level C4B
- Dual adjustable output voltages up to 15V and down to  $-15\text{V}$
- 2A typical switch-current limit for boost and inverter main switches
- High conversion efficiency
  - Up to 91% at positive output rail
  - Up to 85% at negative output rail
  - Power-save mode at low load
- Independent enable inputs for power-up and power-down Sequencing
- Control output for external PFET to support complete supply Disconnect When Shut Down
- 2.7V to 5.5V input-voltage range
- Minimum 1.25MHz fixed-frequency PWM operation
- Thermal shutdown
- Overvoltage protection on both outputs
- $0.2\mu\text{A}$  typical shutdown current
- Small 4mm  $\times$  4mm QFN-24 package (RGE) with wettable flanks

## 2 Applications

- Small-to-medium size OLED displays
- (TFT) LCD, CCD bias supply

## 3 Description

The TPS65131-Q1 device is dual-output dc-dc converter generating a positive output voltage up to 15V and a negative output voltage down to  $-15\text{V}$  with output currents of typically 200mA, depending on input-voltage to output-voltage ratio. With a total efficiency up to 85%, the device is ideal for portable battery-powered equipment. The input-voltage range of 2.7V to 5.5V allows, for example, 3.3V and 5V rails to power the TPS65131-Q1 device. The TPS65131-Q1 device comes in a QFN-24 package with thermal pad and wettable flanks. Requiring few and small external components, the overall solution size can be small.

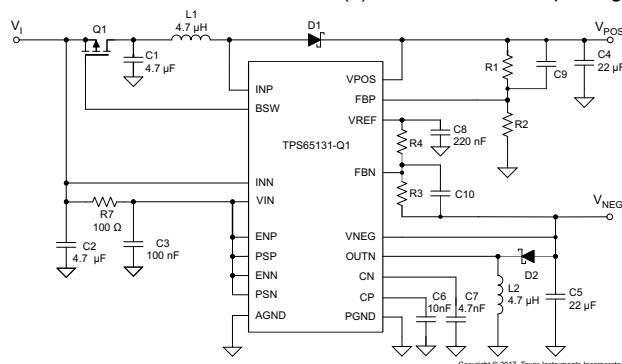
The converter operates with a fixed-frequency PWM control topology and, with power-save mode enabled, uses a pulse-skipping mode at light load currents. In operation, the typical overall device quiescent current is only  $500\mu\text{A}$ . In shutdown, the device draws typically  $0.2\mu\text{A}$ . Independent enable pins allow power-up and power-down sequencing for both outputs. The device has an internal current limit, overvoltage protection, and a thermal shutdown for highest reliability under fault conditions.

The TPS65131-Q1 device is qualified for automotive applications, according to AEC-Q100 temperature grade 2. The electrical characteristics are tested over  $-40^{\circ}\text{C}$  to  $125^{\circ}\text{C}$  device junction temperature. This, combined with lowest shutdown currents, small solution size, package with thermal pad, plus good efficiency and protection features, targets automotive and industrial applications.

### Package Information <sup>(1)</sup>

| PART NUMBER | PACKAGE   | BODY SIZE (NOM)  |
|-------------|-----------|------------------|
| TPS65131-Q1 | VQFN (24) | 4mm $\times$ 4mm |

(1) For all available packages, see the orderable addendum.



## Application Schematic



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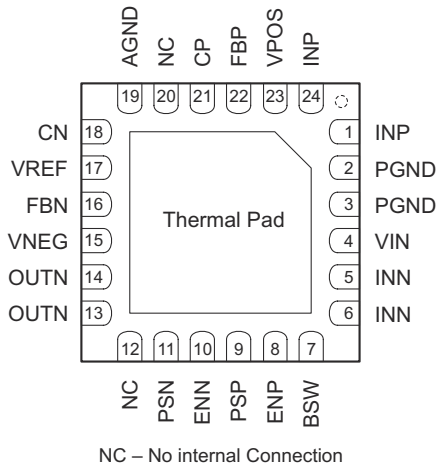
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## 4 Device Comparison

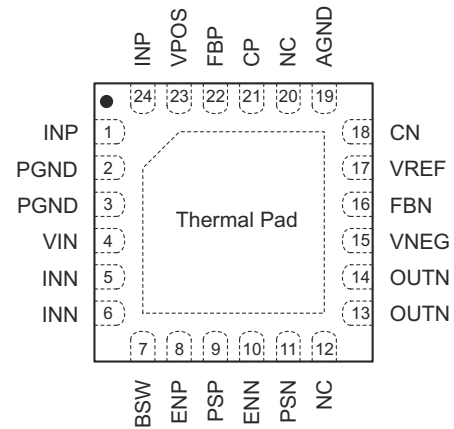
**Table 4-1. Device Comparison Table**

| Part Number      | Package   | Wettable Flanks |
|------------------|-----------|-----------------|
| TPS65131TRGERQ1  | VQFN (24) | No              |
| TPS65131WTRGERQ1 | VQFN (24) | Yes             |

## 5 Pin Configuration and Functions



**Figure 5-1. 24-pin VQFN Bottom View**



**Figure 5-2. 24-pin VQFN Top View**

**Table 5-1. Pin Functions**

| PIN               |        | I/O | DESCRIPTION                                                                                                                                                |
|-------------------|--------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME              | NO.    |     |                                                                                                                                                            |
| AGND              | 19     | —   | Analog ground pin                                                                                                                                          |
| BSW               | 7      | O   | Gate-control pin for external battery switch. This pin goes low when ENP is set high.                                                                      |
| CN                | 18     | I/O | Compensation pin for inverting converter control                                                                                                           |
| CP                | 21     | I/O | Compensation pin for boost converter control                                                                                                               |
| ENN               | 10     | I   | Enable pin for the negative-output voltage (0V: disabled, VIN: enabled)                                                                                    |
| ENP               | 8      | I   | Enable pin for the positive-output voltage (0V: disabled, VIN: enabled)                                                                                    |
| FBN               | 16     | I   | Feedback pin for the negative-output voltage divider                                                                                                       |
| FBP               | 22     | I   | Feedback pin for the positive-output voltage divider                                                                                                       |
| INN               | 5, 6   | O   | Inverting converter switch pin                                                                                                                             |
| INP               | 1, 24  | O   | Boost converter switch pin                                                                                                                                 |
| NC <sup>(1)</sup> | 12, 20 | —   | Not connected                                                                                                                                              |
| OUTN              | 13, 14 | I/O | Inverting converter switch output                                                                                                                          |
| PGND              | 2, 3   | —   | Power ground pin                                                                                                                                           |
| PSN               | 11     | I   | Power-save mode enable for inverter stage (0V: disabled, VIN: enabled)                                                                                     |
| PSP               | 9      | I   | Power-save mode enable for boost converter stage (0V: disabled, VIN: enabled)                                                                              |
| VIN               | 4      | I   | Control supply input                                                                                                                                       |
| VNEG              | 15     | I   | Negative-output voltage-sense input                                                                                                                        |
| VPOS              | 23     | I   | Positive-output voltage-sense input                                                                                                                        |
| VREF              | 17     | O   | Reference output voltage. Bypass this pin with a 220nF capacitor to ground. Connect the lower resistor of the negative-output voltage divider to this pin. |
| Thermal pad       |        |     | Thermal pad for thermal performance, connect to PGND                                                                                                       |

(1) NC - No internal connection

## 6 Specifications

### 6.1 Absolute Maximum Ratings

over operating free-air temperature, unless otherwise noted <sup>(1)</sup>

|                  |                                                                          | VALUE |                   | UNIT |
|------------------|--------------------------------------------------------------------------|-------|-------------------|------|
|                  |                                                                          | MIN   | MAX               |      |
|                  | Input voltage range at pins VIN, INN <sup>(2)</sup>                      | –0.3  | 6                 | V    |
|                  | Voltage at pin VPOS <sup>(2)</sup>                                       | –0.3  | 17                | V    |
|                  | Voltage at pin VNEG <sup>(2)</sup>                                       | –17   | $V_{(VIN)} + 0.3$ | V    |
|                  | Voltage at pins ENN, ENP, FBP, FBN, CN, CP, PSP, PSN, BSW <sup>(2)</sup> | –0.3  | $V_{(VIN)} + 0.3$ | V    |
|                  | Input voltage at pin INP <sup>(2)</sup>                                  | –0.3  | 17                | V    |
|                  | Differential voltage between pins OUTN to INN <sup>(2)</sup>             | –0.3  | 24                | V    |
|                  | Thermal pad <sup>(2)</sup>                                               | –0.3  | 0.3               | V    |
| T <sub>J</sub>   | Operating junction temperature                                           | –40   | 150               | °C   |
| T <sub>stg</sub> | Storage temperature range                                                | –65   | 150               | °C   |

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) All voltage values are with respect to the network ground pin, unless otherwise noted.

### 6.2 ESD Ratings

|                    |                         |                                                         | VALUE | UNIT |
|--------------------|-------------------------|---------------------------------------------------------|-------|------|
| V <sub>(ESD)</sub> | Electrostatic discharge | Human body model (HBM), per AEC Q100-002 <sup>(1)</sup> | ±1000 | V    |
|                    |                         | Charged device model (CDM), per AEC Q100-011            | ±750  | V    |

(1) AEC Q100-002 indicates HBM stressing is done in accordance with the ANSI/ESDA/JEDEC JS-001 specification.

### 6.3 Recommended Operating Conditions

over operating free-air temperature, unless otherwise noted

|                                                          |                                                                          | MIN                  | MAX | UNIT |
|----------------------------------------------------------|--------------------------------------------------------------------------|----------------------|-----|------|
| V <sub>I</sub> , V <sub>(VIN)</sub> , V <sub>(INN)</sub> | Application input voltage range, input voltage range at VIN and INN pins | 2.7                  | 5.5 | V    |
| V <sub>POS</sub>                                         | Adjustable output voltage range for the boost converter                  | V <sub>I</sub> + 0.5 | 15  | V    |
| V <sub>NEG</sub>                                         | Adjustable output voltage range for the inverting converter              | –15                  | –2  | V    |
| V <sub>(ENN)</sub> , V <sub>(ENP)</sub>                  | Enable signals voltage                                                   | 0                    | 5.5 | V    |
| V <sub>(PSN)</sub> , V <sub>(PSP)</sub>                  | Power-save mode enable signals voltage                                   | 0                    | 5.5 | V    |
| T <sub>A</sub>                                           | Operating free-air temperature range <sup>(1)</sup>                      | –40                  | 105 | °C   |
| T <sub>J</sub>                                           | Operating junction temperature range                                     | –40                  | 125 | °C   |

(1) In applications where high power dissipation and/or poor package thermal resistance is present, the maximum ambient temperature may require derating. See [Thermal Information](#) for details.

## 6.4 Thermal Information

| THERMAL METRIC      |                                              | TPS65131-Q1 | UNIT |
|---------------------|----------------------------------------------|-------------|------|
|                     |                                              | RGE PACKAGE |      |
|                     |                                              | 24 PINS     |      |
| $R_{\theta JA}$     | Junction-to-ambient thermal resistance       | 34.1        | °C/W |
| $R_{\theta JCTop}$  | Junction-to-case (top) thermal resistance    | 36.8        | °C/W |
| $R_{\theta JB}$     | Junction-to-board thermal resistance         | 12.2        | °C/W |
| $\Psi_{JT}$         | Junction-to-top characterization parameter   | 0.4         | °C/W |
| $\Psi_{JB}$         | Junction-to-board characterization parameter | 12.3        | °C/W |
| $R_{\theta JCbott}$ | Junction-to-case (bottom) thermal resistance | 2.8         | °C/W |

## 6.5 Electrical Characteristics

This specification applies over the full recommended input voltage range  $V_I = 2.7V$  to  $5.5V$  and over the temperature range  $T_J = -40^{\circ}C$  to  $125^{\circ}C$  unless otherwise noted. Typical values apply for  $V_I = 3.6V$  and  $T_J = 25^{\circ}C$ .

| PARAMETER                                                             |                                              | TEST CONDITIONS                                        | MIN                                                                                         | TYP   | MAX   | UNIT |
|-----------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------|-------|-------|------|
| <b>DC-DC STAGE (<math>V_{(VPOS)}</math>, <math>V_{(VNEG)}</math>)</b> |                                              |                                                        |                                                                                             |       |       |      |
| $V_{ref}$                                                             | Reference voltage                            | $I_{ref} = 10\mu A$                                    | 1.2                                                                                         | 1.213 | 1.225 | V    |
| $I_{(FBP)}$                                                           | Positive feedback input bias current         | $V_{(FBP)} = V_{ref}$                                  |                                                                                             | 50    |       | nA   |
| $I_{(FBN)}$                                                           | Negative feedback input bias current         | $V_{(FBN)} = 0.1V_{ref}$                               |                                                                                             | 50    |       | nA   |
| $V_{(FBP)}$                                                           | Positive feedback regulation voltage         |                                                        | 1.189                                                                                       | 1.213 | 1.237 | V    |
| $V_{(FBN)}$                                                           | Negative feedback regulation voltage         |                                                        | -0.024                                                                                      | 0     | 0.024 | V    |
|                                                                       | Total output dc accuracy                     |                                                        |                                                                                             | 3%    |       |      |
| $r_{DS(on)(N)}$                                                       | Inverter switch on-resistance                | $V_{(VIN)} = 3.6V$                                     |                                                                                             | 440   | 620   | mΩ   |
|                                                                       |                                              | $V_{(VIN)} = 5V$                                       |                                                                                             | 330   | 530   |      |
| $I_{(LIM-N)}$                                                         | Inverter switch current limit                | $V_{(VIN)} = 3.6V$                                     | 1700                                                                                        | 1950  | 2200  | mA   |
| $r_{DS(on)(P)}$                                                       | Boost switch on-resistance                   | $V_{(POS)} = 5V$                                       |                                                                                             | 230   | 390   | mΩ   |
|                                                                       |                                              | $V_{(POS)} = 10V$                                      |                                                                                             | 170   | 230   |      |
| $I_{(LIM-P)}$                                                         | Boost switch current limit                   | $V_{(VIN)} = 3.6V$ , $V_{(POS)} = 8V$                  | 1700                                                                                        | 1950  | 2250  | mA   |
| <b>CONTROL STAGE</b>                                                  |                                              |                                                        |                                                                                             |       |       |      |
| $V_{IH}$                                                              | High-level input voltage, ENP, ENN, PSP, PSN |                                                        | 1.4                                                                                         |       |       | V    |
| $V_{IL}$                                                              | Low-level input voltage, ENP, ENN, PSP, PSN  |                                                        |                                                                                             |       | 0.4   | V    |
|                                                                       | Input current, ENP, ENN, PSP, PSN            | ENP, ENN, PSP, PSN connected to GND or VIN             |                                                                                             | 0.01  | 0.1   | μA   |
| $R_{(BSW)}$                                                           | Output resistance                            |                                                        |                                                                                             | 27    |       | kΩ   |
| $I_Q$                                                                 | Quiescent current                            | VIN                                                    |                                                                                             | 300   | 500   | μA   |
|                                                                       |                                              | VPOS                                                   | $V_{(VIN)} = 3.6V$ , $I_{(POS)} = I_{(NEG)} = 0$ ,<br>ENP = ENN = PSP = PSN = $V_{(VIN)}$ , | 100   | 120   |      |
|                                                                       |                                              | VNEG                                                   | $V_{(POS)} = 8V$ , $V_{(NEG)} = -5V$                                                        | 100   | 120   |      |
| $I_{SD}$                                                              | Shutdown supply current                      | ENN = ENP = LOW, $T_A = -40^{\circ}C$ to $85^{\circ}C$ |                                                                                             | 0.2   | 1.5   | μA   |
| $V_{(UVLO)}$                                                          | Undervoltage lockout threshold               |                                                        | 2.1                                                                                         | 2.35  | 2.7   | V    |
| $T_{(TS)}$                                                            | Thermal shutdown                             |                                                        |                                                                                             | 150   |       | °C   |
| $T_{(TS-HYS)}$                                                        | Thermal shutdown hysteresis                  | Junction temperature decreasing                        |                                                                                             | 5     |       | °C   |

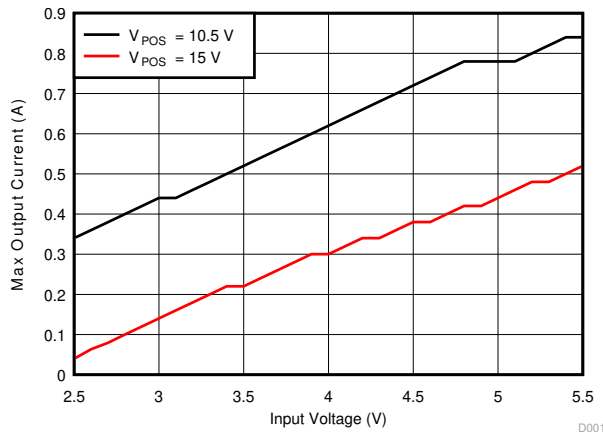
## 6.6 Switching Characteristics

The specification applies over the full recommended input voltage range  $V_I = 2.7\text{V}$  to  $5.5\text{V}$  and over the temperature range  $T_J = -40^\circ\text{C}$  to  $125^\circ\text{C}$  unless otherwise noted. Typical values apply for  $V_I = 3.6\text{V}$  and  $T_J = 25^\circ\text{C}$ .

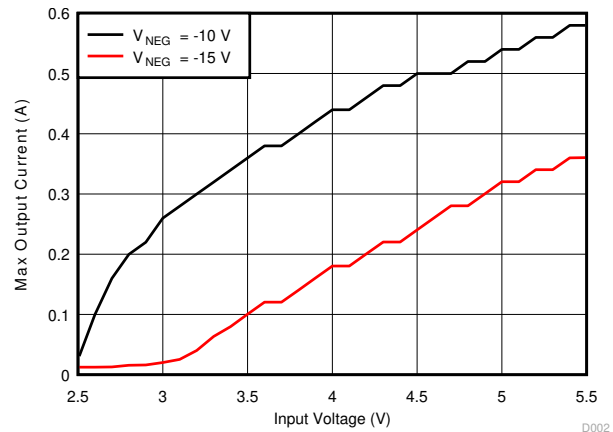
| PARAMETER            | TEST CONDITIONS                         | MIN  | TYP   | MAX  | UNIT |
|----------------------|-----------------------------------------|------|-------|------|------|
| <b>FREQUENCY</b>     |                                         |      |       |      |      |
| f                    | Oscillator frequency                    | 1150 | 1380  | 1500 | kHz  |
| <b>DUTY CYCLE</b>    |                                         |      |       |      |      |
| $D_{(\text{MAX-P})}$ | Maximum-duty-cycle, boost converter     |      | 87.5% |      |      |
| $D_{(\text{MAX-N})}$ | Maximum-duty-cycle, inverting converter |      | 87.5% |      |      |
| $D_{(\text{MIN-P})}$ | Minimum-duty-cycle, boost converter     |      | 12.5% |      |      |
| $D_{(\text{MIN-N})}$ | Minimum-duty-cycle, inverting converter |      | 12.5% |      |      |

## 6.7 Typical Characteristics

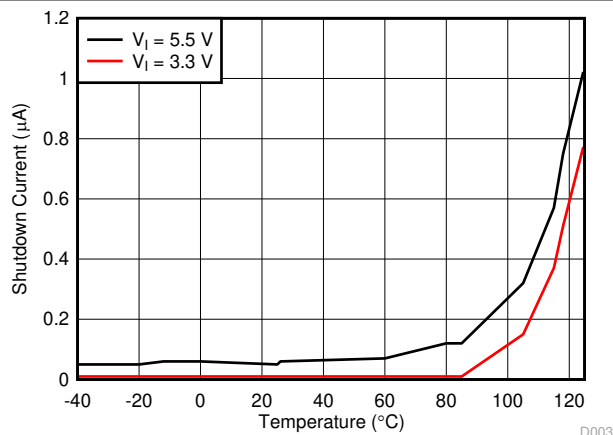
At  $25^\circ\text{C}$ , unless otherwise noted.



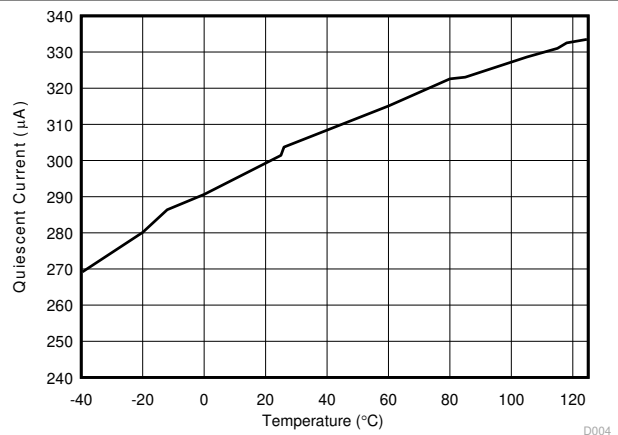
**Figure 6-1. Boost Converter ( $V_{\text{POS}}$ ) Maximum Output Current vs Input Voltage**



**Figure 6-2. Inverting Converter ( $V_{\text{POS}}$ ) Output Current vs Input Voltage**



**Figure 6-3. Shutdown Current (Into VIN and INN) Over Input Voltage**



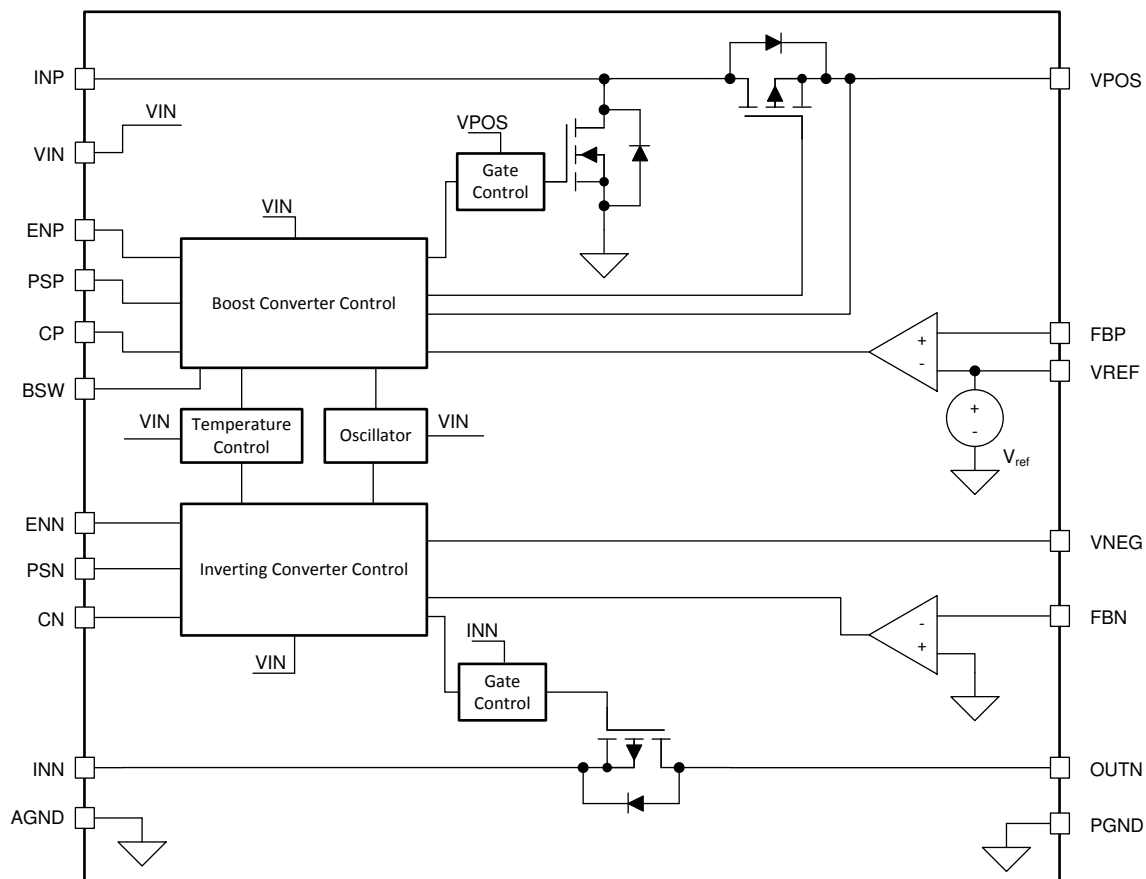
**Figure 6-4. Quiescent Current (Into VIN and INN) Over Input Voltage**

## 7 Detailed Description

### 7.1 Overview

The TPS65131-Q1 is a dual-output dc-dc converter that generates two adjustable output voltages. One output voltage is positive (boost converter), the other is negative (inverting converter). The positive output is adjustable up to 15V, the negative output is adjustable down to -15V. The device operates with an **input voltage range of 2.7V to 5.5V**. Both converters (positive and negative output) work independently of each other. They share a common clock and a common voltage reference. A fixed-frequency, pulse-width-modulated (PWM) regulator controls both outputs separately. In general, each converter operates in continuous-conduction mode (CCM). To improve efficiency at light loads, the converters can operate in discontinuous-conduction mode (DCM). When the power-save mode is enabled, the converters automatically transition between CCM and DCM operation: As the load current decreases, the converter enters DCM mode. Power-save mode is individually configurable for both outputs. The transition as a function of the load current works independently for each converter.

### 7.2 Functional Block Diagram



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## 7.3 Feature Description

### 7.3.1 Power Conversion

Both converters operate in a fixed-frequency, PWM control scheme. The on-time of the internal switches varies depending on the input-to-output voltage ratio and the load. During the on-time, the inductors connected to the converters charge with current. In the remaining time, the off-time with a time period set by the fixed operating frequency, the inductors discharge into the output capacitors through the rectifier diodes. Usually at higher loads, the inductor currents are continuous. At lighter loads, the boost converter uses an additional internal switch to allow current to flow back to the input. This avoids inductor current becoming discontinuous in the boost converter. At the inverting converter, during light loads, the inductor current can become discontinuous. In this case, the control circuit of the inverting controller output automatically takes care of these changing conditions to operate always with an optimum control setup.

### 7.3.2 Control

The controller circuits of both converters employ a fixed-frequency, multiple-feedforward controller topology. These circuits monitor input voltage, output voltage, and voltage drop across the switches. Changes in the operating conditions of the converters directly affect the duty cycle and must not take the indirect and slow way through the output voltage-control loops. A self-learning control corrects measurement errors in this feedforward system. An external capacitor damps the output to avoid output-voltage steps due to output changes of this self-learning control system.

The voltage loops, determined by the error amplifiers, must only handle small signal errors. The error amplifiers feature internal compensation. Their inputs are the feedback voltages on the FBP and FBN pins. The device uses a comparison of these voltages with the internal reference voltage to generate an accurate and stable output voltage.

### 7.3.3 Output Rails Enable or Disable

Both converters can be enabled or disabled individually. Applying a logic HIGH signal at the enable pins (ENP for the boost converter, ENN for the inverting converter) enables the corresponding output. After enabling, internal circuitry, necessary to operate the specific converter, then turns on, followed by the [Soft Start](#).

Applying a low signal at the enable ENP or ENN pin shuts down the corresponding converter. When both enable pins are low, the device enters shutdown mode, where all internal circuitry turns off. The device now consumes shutdown current flowing into the VIN pin. The output loads of the converters can be disconnected from the input, see [Load Disconnect](#).

### 7.3.4 Load Disconnect

The device supports completely disconnecting the load when the converters are disabled. For the inverting converter, the device turns off the internal PMOS switch. If the inverting converter is turned off, no dc current path remains which could discharge the battery or supply.

This is different for the boost converter. The external rectifying diode, together with the boost inductor, form a dc current path which could discharge the battery or supply if any load connects to the output. The device has no internal switch to prevent current from flowing. For this reason, the device offers a PMOS gate control output (BSW) to enable and disable a PMOS switch in this dc current path, ideally directly between the boost inductor and battery. To be able to fully disconnect the battery, the forward direction of the parasitic backgate diode of this switch must point to the battery or supply. The external PMOS switch, which connects to BSW, turns on when the boost converter is enabled and turns off when the boost converter is disabled.



### 7.3.5 Soft Start

Both converters have implemented soft-start functions. When each converter is enabled, the implemented switch current limit ramps up slowly to its nominal programmed value in typically 1ms. The device includes this function to limit the input current during start-up to avoid high peak input currents, which could interfere with other systems connected to the same battery or supply.

If the application includes the [Load Disconnect](#) PMOS switch, a current flows from the input to the output of the boost converter at the moment the PMOS switch becomes conducting.

### 7.3.6 Overvoltage Protection

Both built-in converters (boost and inverter) have implemented individual overvoltage protection. If the feedback voltage under normal operation exceeds the nominal value by typically 5%, the corresponding converter shuts down immediately to protect any connected circuitry from possible damage.

### 7.3.7 Undervoltage Lockout

An undervoltage lockout prevents the device from starting up and operating if the supply voltage at the VIN pin is lower than the undervoltage lockout threshold. For this case, the device automatically shuts down both converters when the supply voltage at VIN falls below this threshold. Nevertheless, parts of the control circuits remain active, which is different than device shutdown using EN inputs. The device includes the undervoltage lockout function to prevent device malfunction.

### 7.3.8 Overtemperature Shutdown

The device automatically shuts down both converters if the implemented internal temperature sensor detects a chip temperature above the thermal shutdown temperature. It automatically starts operating again when the chip temperature falls below this threshold plus hysteresis threshold. The built-in hysteresis avoids undefined operation caused by ringing from shutdown and prevents operating at a temperature close to the overtemperature shutdown threshold.

## 7.4 Device Functional Modes

### 7.4.1 Power-Save Mode

The power-save mode can improve efficiency at light loads. In power-save mode, the converter only operates when the output voltage falls below an device internally set threshold voltage. The converter ramps up the output voltage with one or several operating pulses and goes again into power-save mode once the inductor current becomes discontinuous.

The PSN and PSP logic level selects between power-save mode and continuous-conduction mode. If the specific pins (PSP for the boost converter, PSN for the inverting converter) are HIGH, the power-save mode for the corresponding converter operates at light loads. Similarly, a LOW on the PSP pin or PSN pin disables the power-save mode for the corresponding converter.

## 8 Application and Implementation

### Note

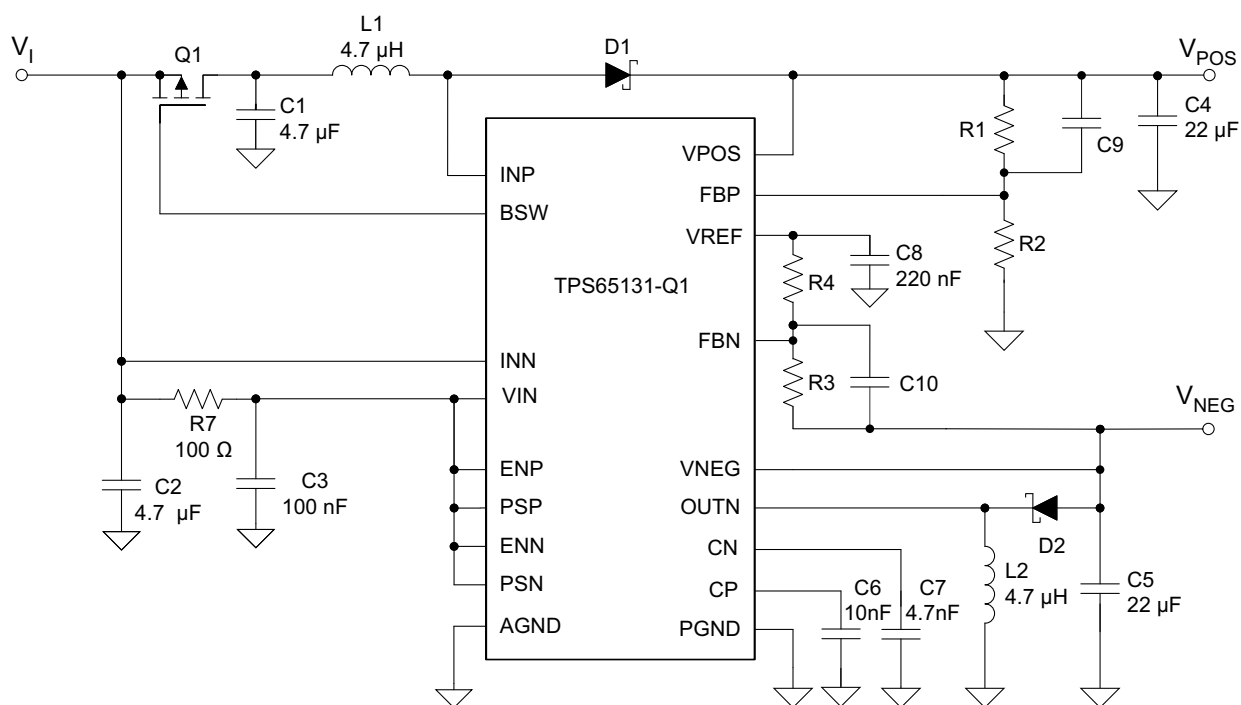
Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

### 8.1 Application Information

The TPS656131-Q1 boost converter output voltage,  $V_{POS}$ , and the inverting converter output voltage,  $V_{NEG}$ , require external components to set the required output voltages. The valid output voltage ranges are as shown in [Recommended Operating Conditions](#) ). The passages below show typical application examples with different output voltage settings and guidance for external component choices.

### 8.2 Typical Applications

#### 8.2.1 TPS65131-Q1 With $V_{POS} = 10.5V$ , $V_{NEG} = -10V$



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Figure 8-1. Typical Application Schematic With  $V_{POS} = 10.5V$ ,  $V_{NEG} = -10V$

## 8.2.2 Design Requirements

The design procedure for this setup is similar to the first example, see [Detailed Design Procedure](#). Change the feedback dividers to set the output voltage, see [Programming the Output Voltage](#). Further, choose the feed-forward capacitors according to [Feedforward Capacitors](#). Table 8-1 shows the components being changed. See Figure 8-1.

**Table 8-1. Design Parameters**

| Design Parameter                                     | Example Value                           |       |
|------------------------------------------------------|-----------------------------------------|-------|
| Input voltage range                                  | 2.7V to 5.5V                            |       |
| Boost converter output voltage, V <sub>POS</sub>     | R1 = 1MΩ<br>R2 = 130kΩ<br>C9 = 6.8pF    | 10.5V |
| Inverting converter output voltage, V <sub>NEG</sub> | R3 = 1MΩ<br>R4 = 121.2kΩ<br>C10 = 7.5pF | –10V  |

In this example, the converters are operated with power-save mode both enabled and disabled (see [Power-Save Mode](#)).

## 8.2.3 Detailed Design Procedure

### 8.2.3.1 Programming the Output Voltage

#### 8.2.3.1.1 Boost Converter

An external resistor divider adjusts the output voltage of the TPS65131-Q1 boost converter stage. Connect this divider to the FBP pin. The typical value of the voltage at the FBP pin is the **reference voltage, which is 1.213V**. The maximum recommended output voltage at the boost converter is 15V. To achieve appropriate accuracy, the current through the feedback divider should be about 100 times higher than the current into the FBP pin. Typical current into the FBP pin is 0.05μA, and the voltage across R2 is 1.213V. Based on those values, **the recommended value for R2 should be lower than 200kΩ** in order to set the divider current at 5μA or higher.

Calculate the value of resistor R1, as a function of the needed output voltage (V<sub>POS</sub>), with [Equation 1](#):

$$R1 = R2 \times \left( \frac{V_{POS}}{V_{ref}} - 1 \right) \quad (1)$$

In this example, with R2 = 130kΩ, choose R1 = 1MΩ to set V<sub>POS</sub> = 10.5V.

#### 8.2.3.1.2 Inverting Converter

An external resistor divider adjusts the output voltage of the TPS65131-Q1 inverting converter stage. Connect this divider to the FBN pin. Unlike the feedback divider at the boost converter, the reference point of the feedback divider is not GND, but V<sub>ref</sub>. So the typical value of the voltage at the FBN pin is 0V. The **minimum recommended output voltage at the inverting converter is –15V**. Feedback divider current considerations are similar to the considerations for the boost converter. For the same reasons, the feedback divider current should be in the range of 5μA or higher. The voltage across R4 is 1.213V. Based on those values, **the recommended value for R4 should be lower than 200kΩ** in order to set the divider current at the required value.

Calculate the value of resistor R3, as a function of the needed output voltage (V<sub>NEG</sub>), with [Equation 2](#):

$$R3 = -R4 \times \left( \frac{V_{NEG}}{V_{ref}} \right) \quad (2)$$

In this example, with R4 = 121.2kΩ, choose R3 = 1MΩ to set V<sub>NEG</sub> = –10V.

### 8.2.3.1.3 Inductor Selection

An inductive converter normally requires two main passive components to store energy during the conversion. Therefore, each converter requires an inductor and a storage capacitor. To select the right inductor, it is recommended to keep the possible peak inductor current below the current-limit threshold of the power switch in the chosen configuration. For example, the current-limit threshold of the switch for the boost converter and for the inverting converters is nominally 1950mA. The highest peak current through the switches and the inductor depends on the output load ( $I_{POS}$ ,  $I_{NEG}$ ), the input voltage ( $V_I$ ), and the output voltages ( $V_{POS}$ ,  $V_{NEG}$ ). Use Equation 3 to estimate the peak inductor current in the boost converter,  $I_{(L-P)}$ . Equation 4 shows the corresponding formula for the inverting converter,  $I_{(L-N)}$ .

$$I_{(L-P)} = \frac{V_{POS}}{V_I \times 0.64} \times I_{POS} \quad (3)$$

$$I_{(L-N)} = \frac{V_I - V_{NEG}}{V_I \times 0.64} \times I_{NEG} \quad (4)$$

The second parameter for choosing the inductor is the desired current ripple in the inductor. Normally, it is advisable to work with a ripple of less than 20% of the average inductor current. A smaller ripple reduces the losses in the inductor, as well as output voltage ripple and EMI. But in the same way, output voltage regulation gets slower, causing higher voltage changes during fast load changes. In addition, a larger inductor usually increases the total system cost. Keep those parameters in mind and calculate the possible inductor value with Equation 5 for the boost converter (L1) and Equation 6 for the inverting converter (L2).

$$L1 = \frac{V_I \times (V_{POS} - V_I)}{\Delta I_{(L-P)} \times f \times V_{POS}} \quad (5)$$

$$L2 = \frac{V_I \times V_{NEG}}{\Delta I_{(L-N)} \times f \times (V_{NEG} - V_I)} \quad (6)$$

The parameter  $f$  is the switching frequency. For the boost converter,  $\Delta I_{(L-P)}$  is the ripple current in the inductor, that is, 20% of  $I_{(L-P)}$ . Accordingly, for the inverting converter,  $\Delta I_{(L-N)}$  is the ripple current in the inductor, that is, 20% of  $I_{(L-N)}$ .  $V_I$  is the input voltage, which is 3.3V in this example. So, the calculated inductance value for the boost inductor is 5.1μH and for the inverting converter inductor is 5.1μH. With these calculated values and the calculated currents, it is possible to choose a suitable inductor.

In typical applications, the recommendation is to choose a 4.7µH inductor. The device is optimized to work with inductance values between 3.3µH and 6.8µH. Nevertheless, operation with higher inductance values may be possible in some applications. Perform detailed stability analysis in this case. Be aware of the possibility that load transients and losses in the circuit can lead to higher currents than estimated in Equation 3 and Equation 4. Also, the losses caused by magnetic hysteresis and conductor resistance are a major parameter for total circuit efficiency.

The following table shows inductors from different suppliers used with the TPS65131-Q1 converter:

**Table 8-2. List of Inductors**

| VENDOR <sup>(1)</sup>           | INDUCTOR SERIES |
|---------------------------------|-----------------|
| EPCOS                           | B8246284-G4     |
| Würth Elektronik                | 7447789XXX      |
|                                 | 744031XXX       |
| TDK                             | VLF3010         |
|                                 | VLF4012         |
| Cooper Electronics Technologies | SD12            |

(1) See Section 9.3

### 8.2.3.2 Capacitor Selection

#### 8.2.3.2.1 Input Capacitor

As a recommendation, choose an input capacitors of at least 4.7µF for the input of the boost converter (INP) and accordingly for the input of the inverting converter (INN). This improves transient behavior of the regulators and EMI behavior of the total power-supply circuit. Choose a ceramic capacitor or a tantalum capacitor. For the use of a tantalum capacitor, an additional, smaller ceramic capacitor (100nF) in parallel is required. Place the input capacitor(s) close to the input pins.



### 8.2.3.2.2 Output Capacitors

One of the major parameters necessary to define the capacitance value of the output capacitor is the maximum allowed output voltage ripple of the converter. Two parameters, which are the capacitance and the equivalent series resistance (ESR), affect this ripple. It is possible to calculate the minimum capacitance needed for the defined ripple, supposing that the ESR is zero. Use [Equation 7](#) for the boost-converter output capacitor (C4min) and [Equation 8](#) for the inverting-converter output capacitor (C5min).

$$C4min = \frac{I_{POS} \times (V_{POS} - V_I)}{f \times \Delta V_{POS} \times V_{POS}} \quad (7)$$

$$C5min = \frac{I_{NEG} \times V_{NEG}}{f \times \Delta V_{NEG} \times (V_{NEG} - V_I)} \quad (8)$$

The parameter  $f$  is the switching frequency.  $\Delta V_{POS}$  and  $\Delta V_{NEG}$  are the maximum allowed ripple voltages for each converter.

Choosing a ripple voltage in the range of 10mV requires a **minimum capacitance of 12μF**. The total ripple is larger due to the ESR of the output capacitor. Use [Equation 9](#) for the boost converter and [Equation 10](#) for the inverting converter to calculate this additional ripple component.

$$\Delta V_{(ESR-P)} = I_{POS} \times R_{(ESR-C4)} \quad (9)$$

$$\Delta V_{(ESR-N)} = I_{NEG} \times R_{(ESR-C5)} \quad (10)$$

In this example, an additional ripple of 2mV is the result of using a typical ceramic capacitor with an ESR in the 10mΩ range. The total ripple is the sum of the ripple caused by the capacitance and the ripple caused by the ESR of the capacitor. In this example, the total ripple is 10mV.

Load transients can create additional ripple. When the load current increases rapidly, the output capacitor must provide the additional current until the inductor current increases by the control loop which sets a higher on-time (duty cycle) of the main switch. The higher duty cycle results in longer inductor charging periods. The inductance itself also limits the rate of increase of the inductor current. When the load current decreases rapidly, the output capacitor must store the excess energy (stored in the inductor) until the regulator has decreased the inductor current by reducing the duty cycle. The recommendation is to use higher capacitance values, as the foregoing calculations show.

### 8.2.3.3 Rectifier Diode Selection

Both converters (the boost and inverting converter) require rectifier diodes, **D1 and D2**. As a recommendation, to reduce losses, use **Schottky diodes**. The **forward current rating needed is equal to the maximum output current**. Consider that the maximum currents,  $I_{POSmax}$  and  $I_{NEGmax}$ , might differ for  $V_{POS}$  and  $V_{NEG}$  when choosing the diodes.

### 8.2.3.4 External P-MOSFET Selection

During shutdown, when connected to a power supply, a path from the power supply to the positive output conducts through the inductor and an external diode. Optionally, in order to fully disconnect the positive output  $V_{POS}$  during shutdown, add an external p-MOSFET (Q1). The BSW pin controls the gate of the p-MOSFET. When choosing a proper p-MOSFET, the  $V_{GS}$  and  $V_{GD}$  voltage ratings must cover the input voltage range, the drain current rating must not be lower than the maximum input current flowing into the application, and conditions of the p-MOSFET operating area must fit.

If there is no intention to use an external p-MOSFET, leave the BSW pin floating.

### 8.2.3.5 Stabilizing the Control Loop

#### 8.2.3.5.1 Feedforward Capacitors

As a recommendation, to speed up the control loop, place feedforward capacitors in the feedback divider, parallel to R1 (boost converter) and R3 (inverting converter). Equation 11 shows how to calculate the appropriate value for the boost converter, and Equation 12 for the inverting converter.

$$C9 = \frac{6.8 \mu s}{R1} \quad (11)$$

$$C10 = \frac{7.5 \mu s}{R3} \quad (12)$$

In this application example,  $C9 = 6.8pF$  and  $C10 = 7.5pF$  match the choices of R1 and R3.

To avoid coupling noise into the control loop from the feedforward capacitors, it is possible to place a series resistor to limit the bandwidth of the feedforward effect. Any value between 10kΩ and 100kΩ is suitable. The higher the resistance, the lower the noise coupled into the control loop system.

#### 8.2.3.5.2 Compensation Capacitors

The device features completely internally compensated control loops for both converters. The internal feedforward system has built-in error correction which requires external capacitors. As a recommendation, use a 10nF capacitor at the CP pin of the boost converter and a 4.7nF capacitor at the CN pin of the inverting converter.

### 8.2.4 Analog Supply Input Filter

To ensure a noise free voltage supply of the IC, it is recommended to add an RC or LC filter between IIN and VIN pins.

#### 8.2.4.1 RC-Filter

For most applications an RC filter can be used with a resistance value of 100Ω minimum and capacitor value of 0.1μF as in the application example [Figure 8-1](#).

#### 8.2.4.2 LC-Filter

For applications where input voltages  $V_I$  with a fast rising edge (slew rate  $\geq 275\text{mV}/\mu\text{s}$ ) are expected, it is recommended to replace the resistor R7 with a ferrite bead to minimize the delay between the signals on IIN and VIN. A ferrite bead with the lowest possible DCR and a proper current rating should be selected - BLM18KG101TN1 for example. A conservative approach for the current rating specification is to set it at 1.5 times or twice the maximum input current.

**Table 8-3. List of Ferrite Beads**

| VENDOR | FERRITE BEAD SERIES |
|--------|---------------------|
| Murata | BLMxKG              |

### 8.2.5 Thermal Information

Implementation of integrated circuits in low-profile and fine-pitch surface-mount packages typically requires special attention to power dissipation. Many system-dependent issues, such as thermal coupling, airflow, added heatsinks and convection surfaces, and the presence of heat-generating components affect the power-dissipation limits of a given component.

Three basic approaches for enhancing thermal performance follow.

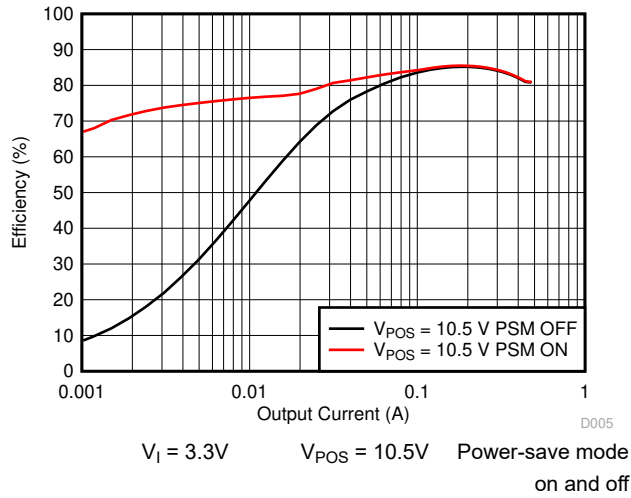
- Improving the power dissipation capability of the PCB design
- Improving the thermal coupling of the component to the PCB
- Introducing airflow to the system

The recommended device junction temperature range,  $T_J$ , is  $-40^\circ\text{C}$  to  $125^\circ\text{C}$ . The thermal resistance of the 24-pin QFN, 4-mm × 4-mm package (RGE) is  $R_{\theta JA} = 34.1^\circ\text{C}/\text{W}$ . The recommended operating ambient temperature range for the device is  $T_A = -40^\circ\text{C}$  to  $105^\circ\text{C}$ . Use [Equation 13](#) to calculate the maximum power dissipation,  $P_{D\text{max}}$ , as a function of  $T_A$ . In this equation, use  $T_J = 125^\circ\text{C}$  to operate the device within the recommended temperature range, use  $T_J = T_{(TS)}$  to determine the absolute maximum threshold when the device might go into thermal shutdown. If the maximum ambient temperature of the application is lower, more heat dissipation is possible.

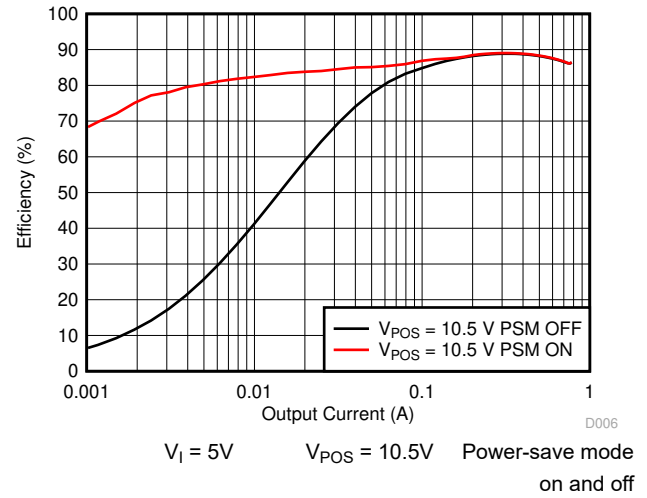
$$P_{D\text{max}} = \frac{T_J - T_A}{R_{\theta JA}} \quad (13)$$



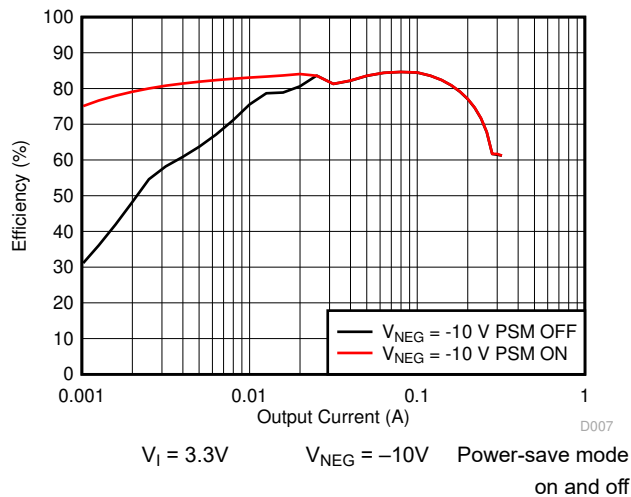
## 8.2.6 Application Curves



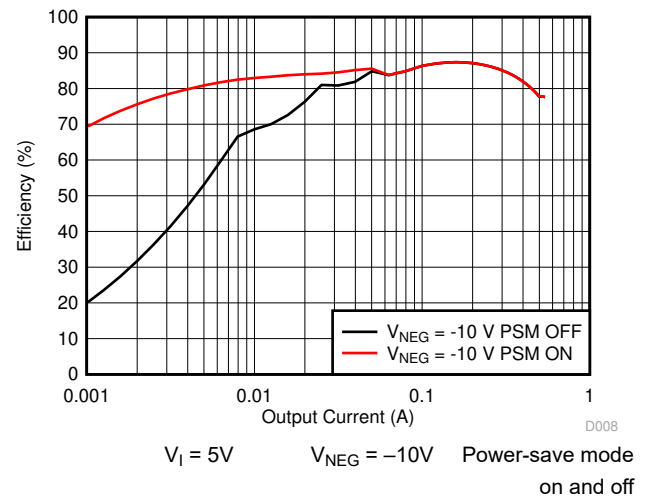
**Figure 8-2. Boost Converter ( $V_{POS}$ ) Efficiency vs Output Current**



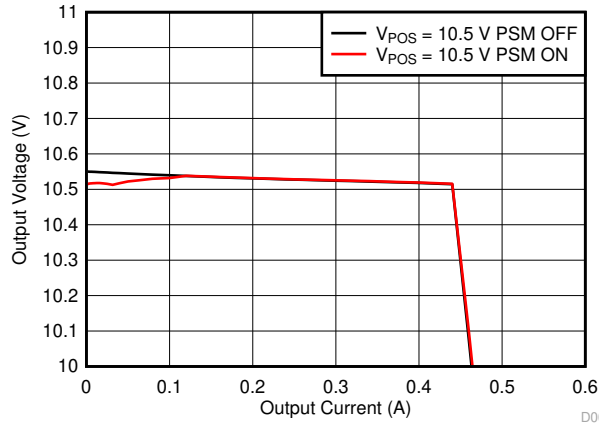
**Figure 8-3. Boost Converter ( $V_{POS}$ ) Efficiency vs Output Current**



**Figure 8-4. Inverting Converter ( $V_{NEG}$ ) Efficiency vs Output Current**

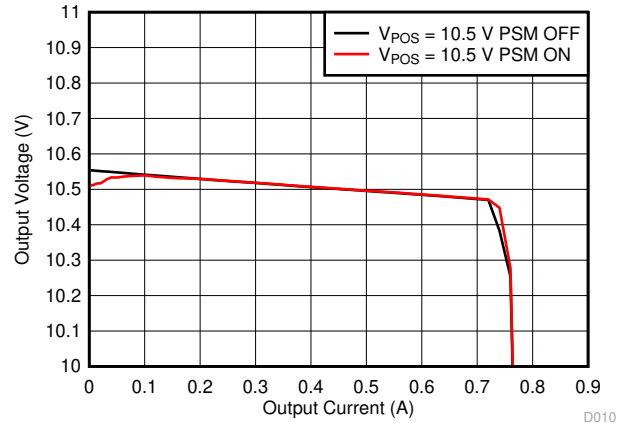


**Figure 8-5. Inverting Converter ( $V_{NEG}$ ) Efficiency vs Output Current**



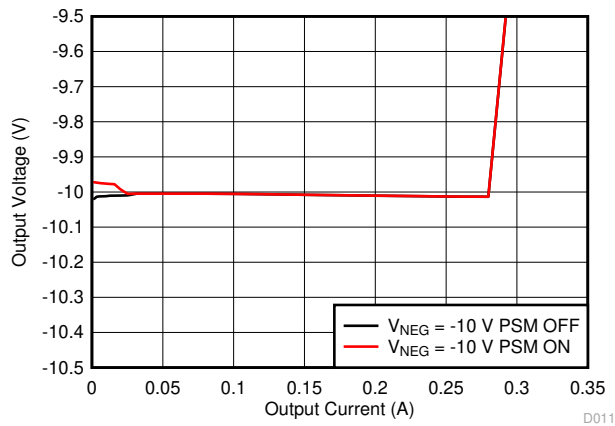
$V_I = 3.3V$   $V_{POS} = 10.5V$  Power-save mode  
on and off

**Figure 8-6. Boost Converter ( $V_{POS}$ ) Output Voltage vs Output Current**



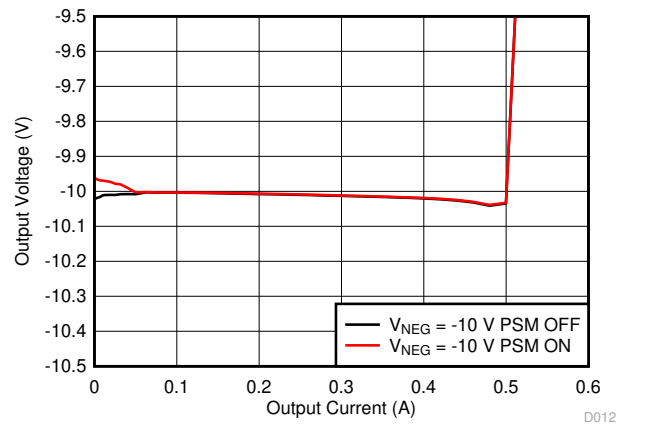
$V_I = 5V$   $V_{POS} = 10.5V$  Power-save mode  
on and off

**Figure 8-7. Boost Converter ( $V_{POS}$ ) Output Voltage vs Output Current**



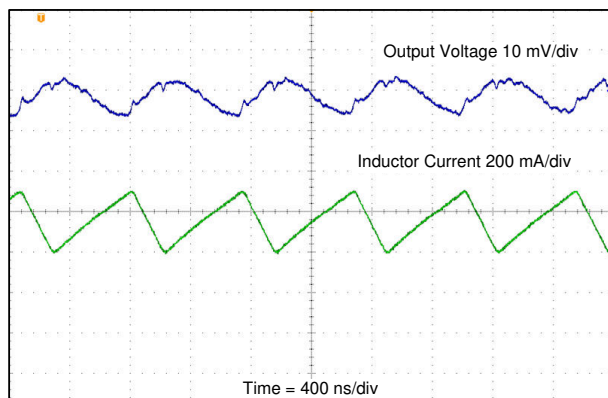
$V_I = 3.3V$   $V_{NEG} = -10V$  Power-save mode  
on and off

**Figure 8-8. Inverting Converter ( $V_{NEG}$ ) Output Voltage vs Output Current**



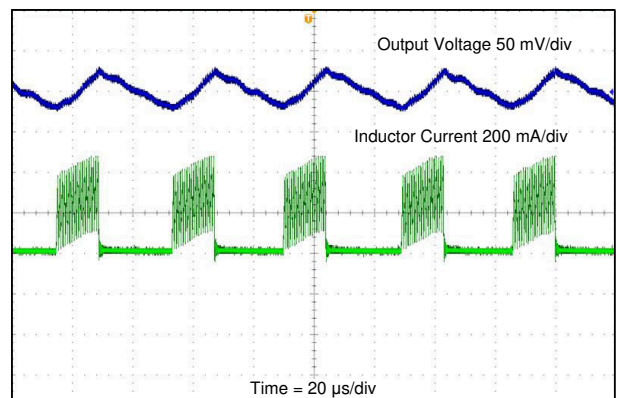
$V_I = 5V$   $V_{NEG} = -10V$  Power-save mode  
on and off

**Figure 8-9. Inverting Converter ( $V_{NEG}$ ) Output Voltage vs Output Current**



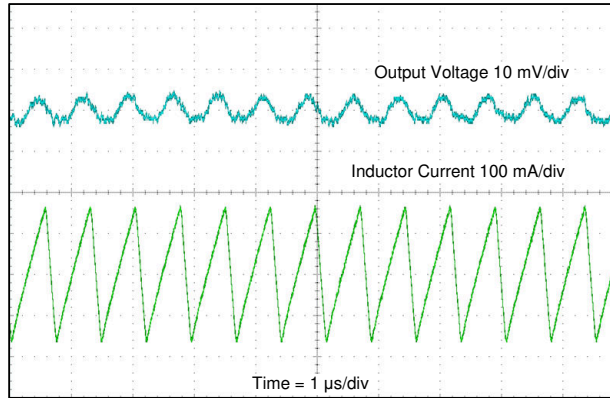
$V_I = 3.3V$   $V_{POS} = 10.5V$   $I_{POS} = 200mA$  Power-save mode  
off

**Figure 8-10. Boost Converter ( $V_{POS}$ ) Output Ripple**

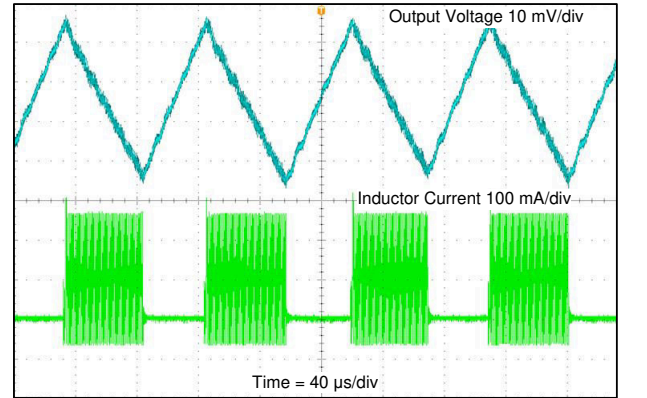


$V_I = 3.3V$   $V_{POS} = 10.5V$   $I_{POS} = 20mA$  Power-save mode  
on

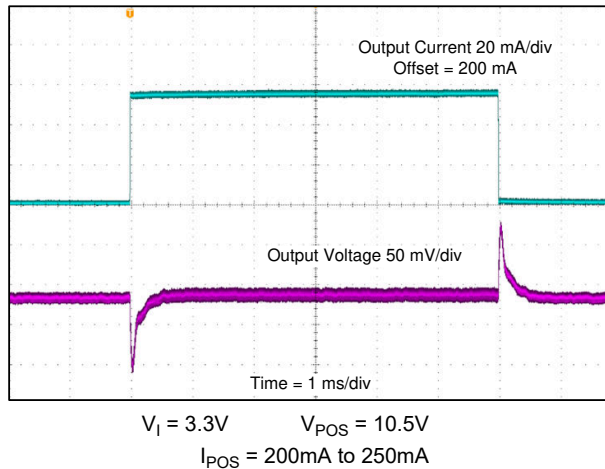
**Figure 8-11. Boost Converter ( $V_{POS}$ ) Output Ripple**



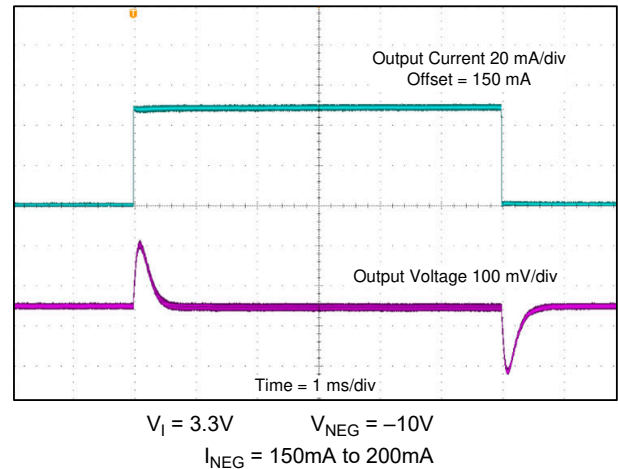
**Figure 8-12. Inverting Converter ( $V_{NEG}$ ) Output Ripple**



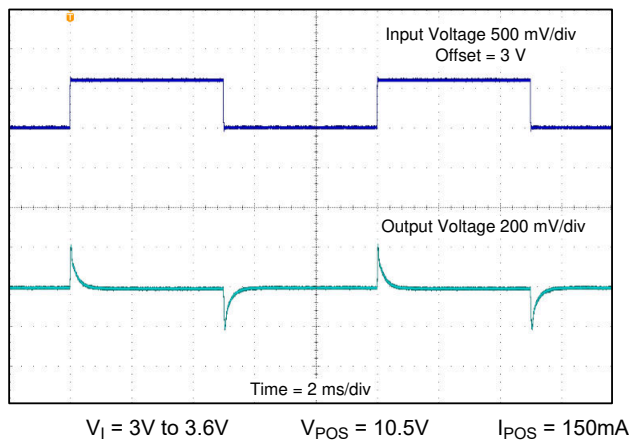
**Figure 8-13. Inverting Converter ( $V_{NEG}$ ) Output Ripple**



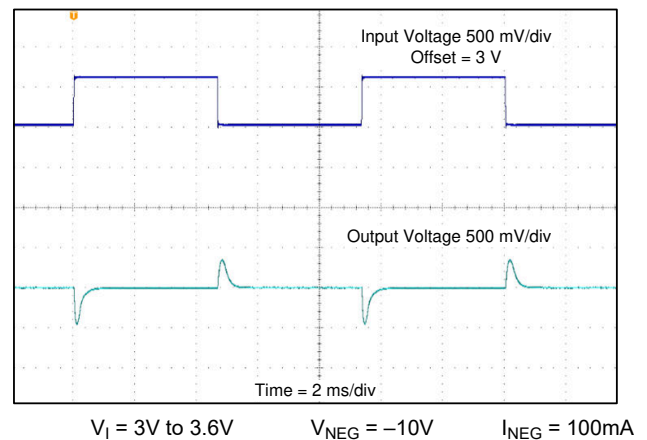
**Figure 8-14. Boost Converter ( $V_{POS}$ ) Load Transient Response**



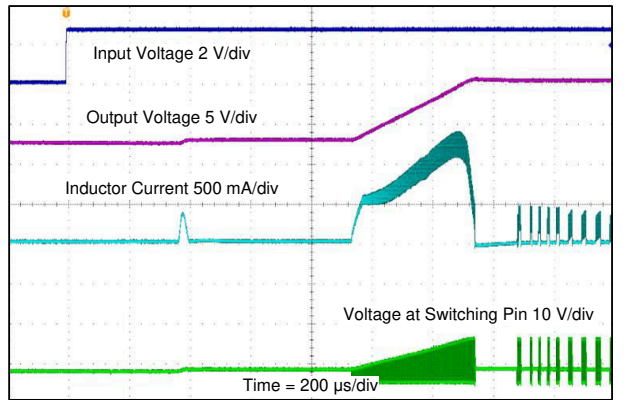
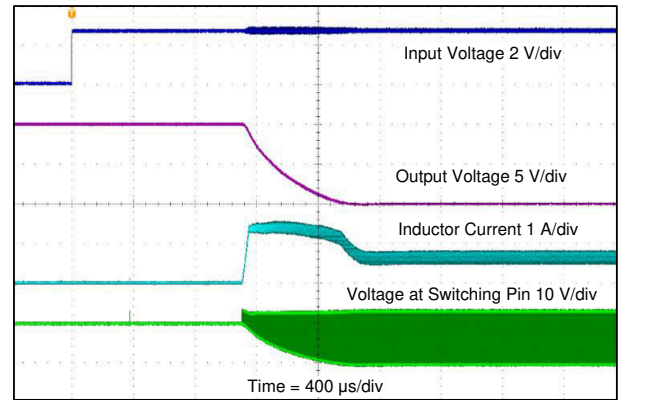
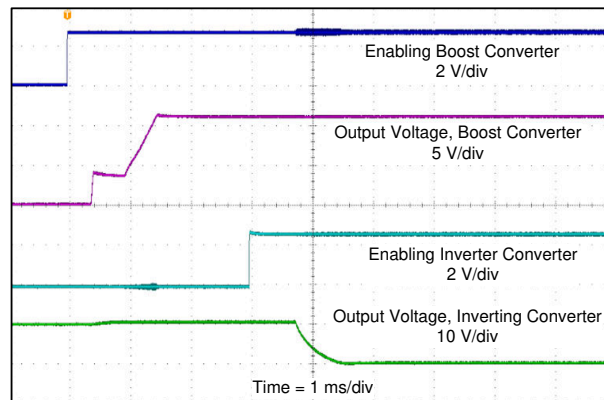
**Figure 8-15. Inverting Converter ( $V_{NEG}$ ) Load Transient Response**



**Figure 8-16. Boost Converter ( $V_{POS}$ ) Line Transient Response**



**Figure 8-17. Inverting ( $V_{NEG}$ ) Converter Line Transient Response**


 $V_I = 3.3V$      $V_{POS} = 10.5V$      $I_{POS} = 46mA$ 
**Figure 8-18. Boost Converter ( $V_{POS}$ ) Start-Up Into Load**

 $V_I = 3.3V$      $V_{NEG} = -10V$      $I_{NEG} = 150mA$ 
**Figure 8-19. Inverting Converter ( $V_{NEG}$ ) Start-Up Into Load**

 $V_I = 3.3V$   
 $I_{POS} = I_{NEG} = 160mA$ 
 $V_{POS} = 10.5V$ 
 $V_{NEG} = -10V$ 
**Figure 8-20. Boost and Inverting Converter Start-Up Into Load**

### 8.3 Power Supply Recommendations

The TPS65131-Q1 input voltage ranges from 2.7V to 5.5V. Consequently, the supply can come, for example, from a 3.3V or 5V rail. If the device starts into load during the [Soft Start](#) phase, the drawn input current can be higher than during post-start operation. Consider the application requirements when selecting the power supply.

To avoid unintended toggling of the [Undervoltage Lockout](#), connect the TPS65131-Q1 via a low-impedance path to the power supply.

## 8.4 Layout

### 8.4.1 Layout Guidelines

As for all switching power supplies, the layout is an important step in the design, especially at high peak currents and high switching frequencies. Improper layout might show the symptoms of poor line or load regulation, ground and output voltage shifts, stability issues, unsatisfying EMI behavior or worsened efficiency. Therefore, use wide and short traces for the main current paths and for the power ground tracks. The input capacitors (C1, C2, C3), output capacitors (C4, C5), the inductors (L1, L2), and the rectifying diodes (D1, D2) should be placed as close as possible to the IC to keep parasitic inductances low. Use a wide PGND plane. Connect the analog ground pin (AGND) to the PGND plane. Further, connect the PGND plane with the exposed thermal pad. Place the feedback dividers as close as possible to the control pin (boost converter) or the VREF pin (inverting converter) of the IC.

Figure 8-21 provides an layout example which is recommended to be followed.

### 8.4.2 Layout Example

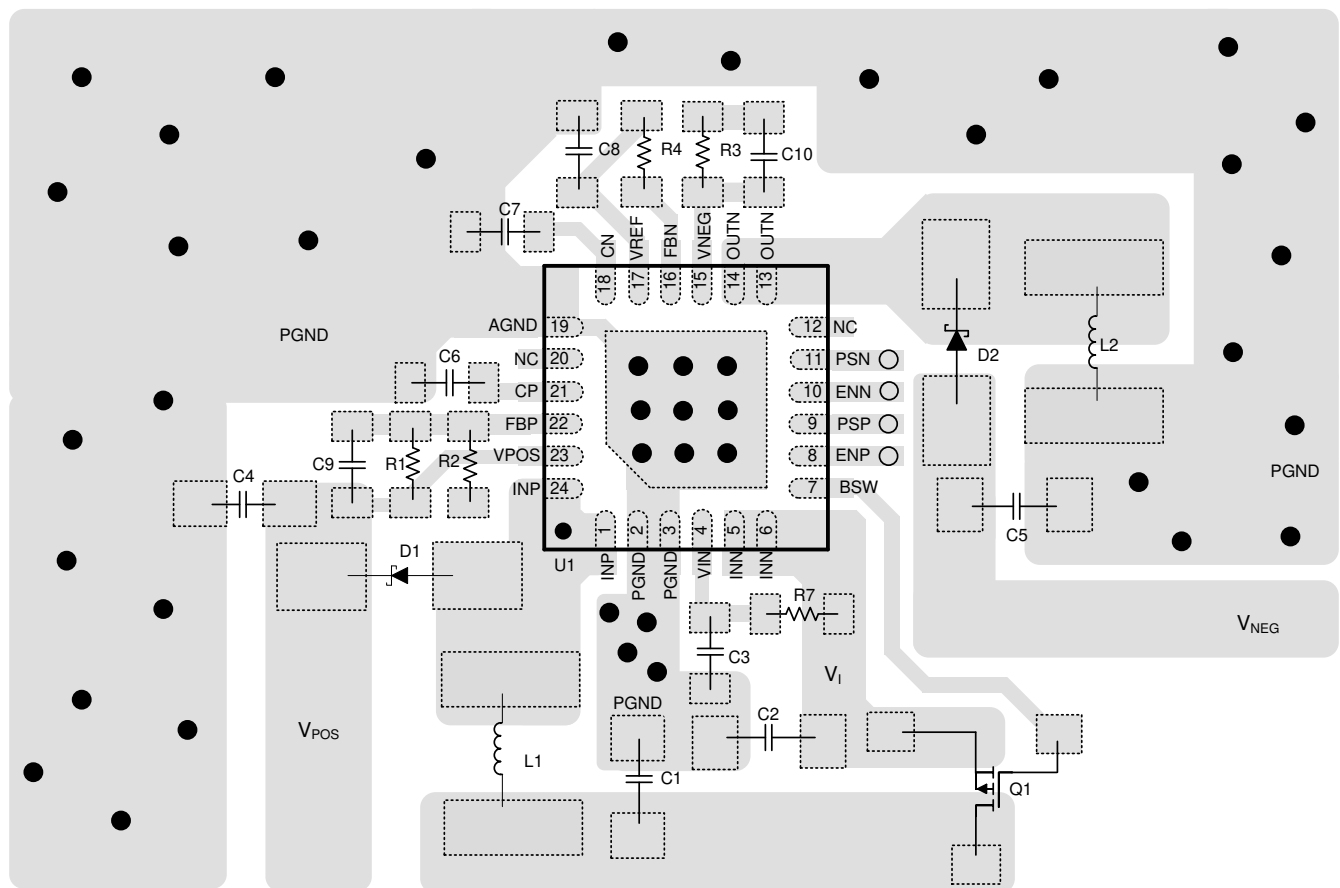


Figure 8-21. TPS65131-Q1 Layout Recommendation

## 9 Device and Documentation Support

TI offers an extensive line of development tools. Tools and software to evaluate the performance of the device, generate code, and develop solutions are listed below.

### 9.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on [ti.com](http://ti.com). Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

### 9.2 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

### 9.3 Trademarks

TI E2E™ is a trademark of Texas Instruments.

All trademarks are the property of their respective owners.

### 9.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

### 9.5 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

## 10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Revision E (March 2017) to Revision F (August 2024) | Page |
|------------------------------------------------------------------|------|
| • Updated <i>Device Information</i> Table.....                   | 1    |

| Changes from Revision D (October 2014) to Revision E (March 2017)                                                                                                                                                                                                                                                                                | Page |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| • Changed <a href="#">Section 1</a> bullet text from "...Qualified.." to...Test Guidance.." and HBM classification level from "H2" to "H1C".....                                                                                                                                                                                                 | 1    |
| • Moved $T_{stg}$ spec to the Abs Max Ratings table per new data sheet standard.....                                                                                                                                                                                                                                                             | 4    |
| • Changed "Handling Ratings" to "ESD Ratings" and HBM Value From " $\pm 2$ kV" to " $\pm 1000$ V".....                                                                                                                                                                                                                                           | 4    |
| • Changed Electrical Characteristics condition statement to "This specification applies over the full recommended input voltage range $V_I = 2.7$ V to 5.5 V and over the temperature range $T_J = -40^\circ\text{C}$ to $125^\circ\text{C}$ unless otherwise noted. Typical values apply for $V_I = 3.6$ V and $T_J = 25^\circ\text{C}$ ."..... | 5    |
| • Changed The specification applies over the full recommended input voltage range $V_I = 2.7$ V to 5.5 V and over the temperature range $T_J = -40^\circ\text{C}$ to $125^\circ\text{C}$ unless otherwise noted. Typical values apply for $V_I = 3.6$ V and $T_J = 25^\circ$ .....                                                               | 6    |
| • Added <a href="#">Section 8.2.4</a> description .....                                                                                                                                                                                                                                                                                          | 16   |



| <b>Changes from Revision C (March 2014) to Revision D (October 2014)</b>                                                                               | <b>Page</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| • Global editorial changes bringing the datasheet into the new format.....                                                                             | 1           |
| • Changed max. efficiency from 89% to 91% and from 81% to 85% .....                                                                                    | 1           |
| • Deleted "Minimum 1.25 MHz".....                                                                                                                      | 1           |
| • Changed 1- $\mu$ A shutdown current to typ. 0.2 $\mu$ A.....                                                                                         | 1           |
| • Added Thermal Pad to Absolute Maximum Ratings. Added min./max. values where missing.....                                                             | 4           |
| • Added $V_{(VIN)}$ , $V_{(INN)}$ , $V_{(NEG)}$ , $V_{(POS)}$ , $V_{(ENN)}$ , $V_{(ENP)}$ , $V_{(PSN)}$ to Recommended Operating Conditions table..... | 4           |
| • Changed symbol names to JEDEC compliance.....                                                                                                        | 5           |
| • Added frequency and duty cycles to Switching Characteristics table. Removed from Electrical Characteristics table.....                               | 6           |
| • Added Rectifier Diode Selection Guide.....                                                                                                           | 15          |
| • Added P-MOSFET Selection Guide.....                                                                                                                  | 15          |

| <b>Changes from Revision B (February 2013) to Revision C (March 2014)</b>                                                                                                                                                                                                                                                                                                                                                                      | <b>Page</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| • Added "Electrical Characteristics tested over $-40^{\circ}\text{C}$ to $125^{\circ}\text{C}$ Junction Temperature Range".....                                                                                                                                                                                                                                                                                                                | 1           |
| • Added Device Information table .....                                                                                                                                                                                                                                                                                                                                                                                                         | 1           |
| • Deleted $T_A$ table row.....                                                                                                                                                                                                                                                                                                                                                                                                                 | 4           |
| • Changed $I_{NN}$ to $V_{INN}$ , added pin names VIN and INN.....                                                                                                                                                                                                                                                                                                                                                                             | 4           |
| • Added pin name VPOS.....                                                                                                                                                                                                                                                                                                                                                                                                                     | 4           |
| • Added pin name VNEG.....                                                                                                                                                                                                                                                                                                                                                                                                                     | 4           |
| • Changed $I_{NP}$ to $V_{INP}$ , added pin name INP.....                                                                                                                                                                                                                                                                                                                                                                                      | 4           |
| • Changed "between pins OUTN to $V_{INN}$ " to "between pins OUTN to INN".....                                                                                                                                                                                                                                                                                                                                                                 | 4           |
| • Added operating junction temperature.....                                                                                                                                                                                                                                                                                                                                                                                                    | 4           |
| • Added "In applications where high power dissipation and/or poor package thermal resistance is present, the maximum ambient temperature may require derating. See <a href="#">Section 8.2.5</a> for details.".....                                                                                                                                                                                                                            | 4           |
| • Deleted "virtual" from "Operating virtual junction temperature range".....                                                                                                                                                                                                                                                                                                                                                                   | 4           |
| • Changed Electrical Characteristics condition statement to "This specification applies over the full recommended input voltage range $V_I = 2.7\text{ V}$ to $5.5\text{ V}$ and over the temperature range $T_J = T_A = -40^{\circ}\text{C}$ to $125^{\circ}\text{C}$ unless otherwise noted. Typical values apply for $V_I = 3.6\text{ V}$ and $T_J = T_A = 25^{\circ}\text{C}$ .".....                                                      | 5           |
| • Changed $I_{LIM,min} = 1800\text{ mA}$ to $1700\text{ mA}$ .....                                                                                                                                                                                                                                                                                                                                                                             | 5           |
| • Deleted $V_{POS} = 5\text{ V}$ ( $105^{\circ}\text{C}$ ) row.....                                                                                                                                                                                                                                                                                                                                                                            | 5           |
| • Changed $r_{DS(on)P,max}$ ( $V_{POS} = 5\text{ V}$ ) = $300\text{ m}\Omega$ to $390\text{ m}\Omega$ .....                                                                                                                                                                                                                                                                                                                                    | 5           |
| • Changed $r_{DS(on)P,max}$ ( $V_{POS} = 10\text{ V}$ ) = $200\text{ m}\Omega$ to $230\text{ m}\Omega$ .....                                                                                                                                                                                                                                                                                                                                   | 5           |
| • Changed $I_{LIMP,min} = 1800\text{ mA}$ to $1700\text{ mA}$ .....                                                                                                                                                                                                                                                                                                                                                                            | 5           |
| • Changed $I_{LIMP,max} = 2200\text{ mA}$ to $2250\text{ mA}$ .....                                                                                                                                                                                                                                                                                                                                                                            | 5           |
| • Added $T_A = -40^{\circ}\text{C}$ to $85^{\circ}\text{C}$ .....                                                                                                                                                                                                                                                                                                                                                                              | 5           |
| • Changed minimum $f = 1250\text{ kHz}$ to $1150\text{ kHz}$ .....                                                                                                                                                                                                                                                                                                                                                                             | 6           |
| • Editorially updated Block Diagram.....                                                                                                                                                                                                                                                                                                                                                                                                       | 7           |
| • Changed "The maximum recommended junction temperature ( $T_J$ ) of the TPS65131-Q1 is $125^{\circ}\text{C}$ ." to "The recommended device junction temperature range, $T_J$ , is $-40^{\circ}\text{C}$ to $125^{\circ}\text{C}$ .".....                                                                                                                                                                                                      | 16          |
| • Changed $R_{\theta JA} = 37.8^{\circ}\text{C/W}$ to $R_{\theta JA} = 34.1^{\circ}\text{C/W}$ .....                                                                                                                                                                                                                                                                                                                                           | 16          |
| • Changed "Specified regulator operation is ensured to a maximum ambient temperature $T_A$ of $105^{\circ}\text{C}$ ." to "The recommended operating ambient temperature range for the device is $T_A = -40^{\circ}\text{C}$ to $105^{\circ}\text{C}$ .".....                                                                                                                                                                                  | 16          |
| • Changed "Therefore, the maximum power dissipation is about $1058\text{ mW}$ " to "Use <a href="#">Equation 13</a> to calculate the maximum power dissipation, $P_{Dmax}$ , as a function of $T_A$ . In this equation, use $T_J = 125^{\circ}\text{C}$ to operate the device within the recommended temperature range, use $T_J = T_{(TS)}$ to determine the absolute maximum threshold when the device might go into thermal shutdown."..... | 16          |
| • Changed <a href="#">Equation 13</a> .....                                                                                                                                                                                                                                                                                                                                                                                                    | 16          |

| <b>Changes from Revision A (November 2012) to Revision B (February 2013)</b> | <b>Page</b>       |
|------------------------------------------------------------------------------|-------------------|
| • Changed CDM ESD rating from C3B to C4B.....                                | <a href="#">1</a> |

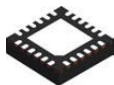
| <b>Changes from Revision * (May 2012) to Revision A (November 2012)</b>                        | <b>Page</b>       |
|------------------------------------------------------------------------------------------------|-------------------|
| • Device is going from Preview to Production.....                                              | <a href="#">1</a> |
| • Added thermal information table values.....                                                  | <a href="#">5</a> |
| • Added $V_{POS} = 5\text{ V}$ (105°C) row and values to Electrical Characteristics table..... | <a href="#">5</a> |

## 11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most-current data available for the designated device. This data is subject to change without notice and without revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.



## 11.1 Mechanical Data

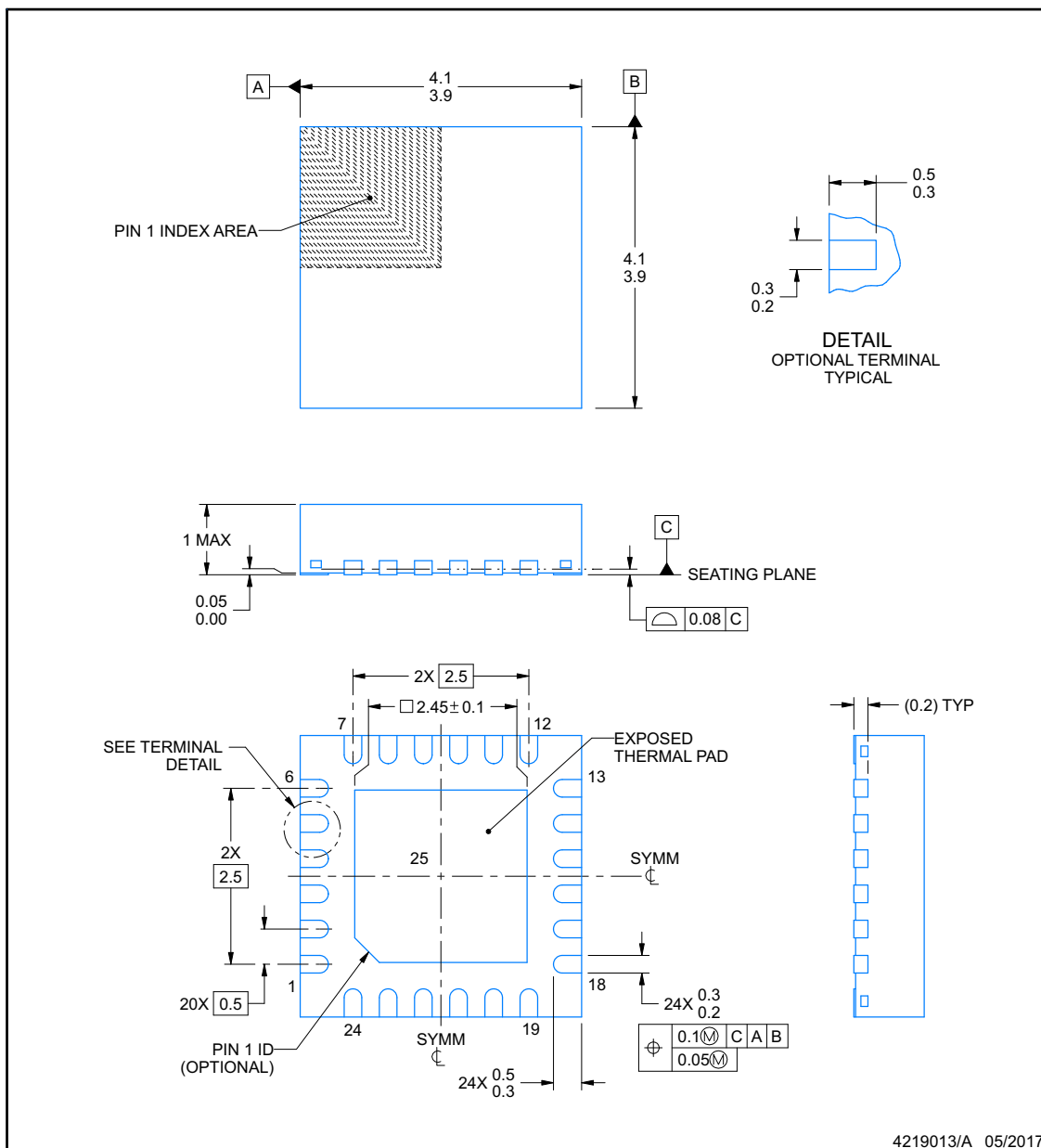


**RGE0024B**

### PACKAGE OUTLINE

**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK - NO LEAD

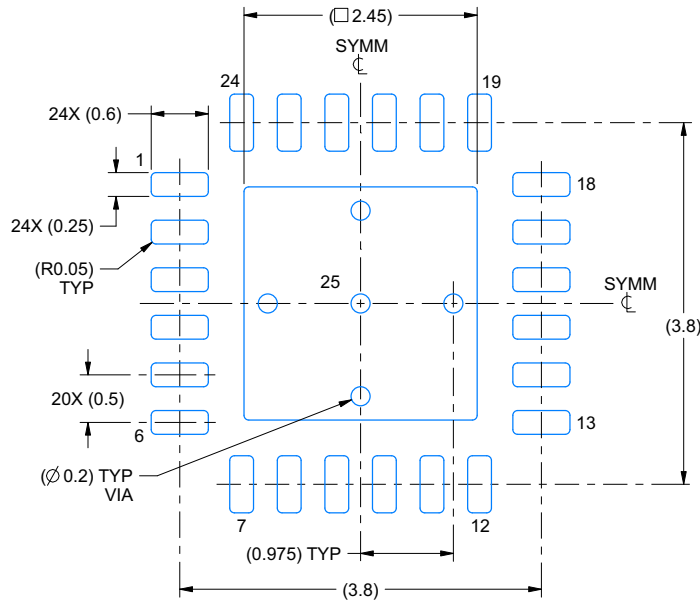


#### NOTES:

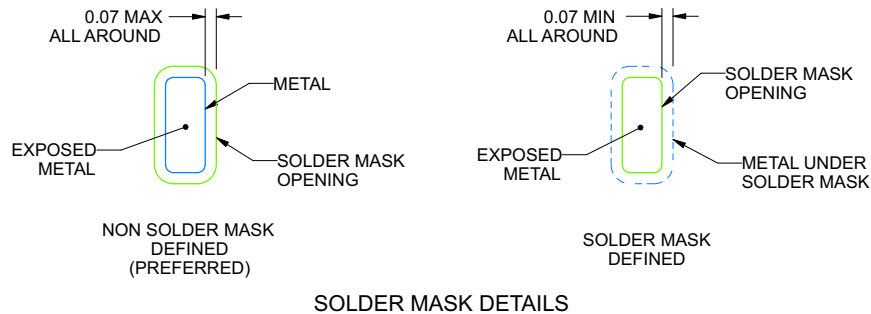
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for thermal and mechanical performance.

**EXAMPLE BOARD LAYOUT****RGE0024B****VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK - NO LEAD



**LAND PATTERN EXAMPLE**  
EXPOSED METAL SHOWN  
SCALE:15X



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NOTES: (continued)

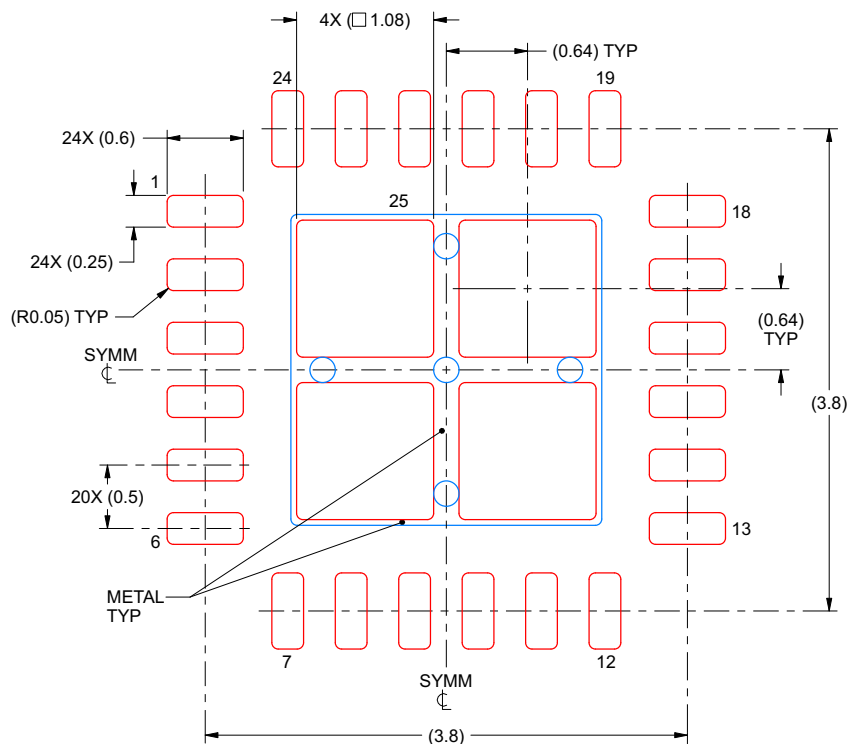
4. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 ([www.ti.com/lit/slua271](http://www.ti.com/lit/slua271)).
5. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

## EXAMPLE STENCIL DESIGN

**RGE0024B**

**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK - NO LEAD



**SOLDER PASTE EXAMPLE**  
BASED ON 0.125 mm THICK STENCIL  
EXPOSED PAD 25  
78% PRINTED SOLDER COVERAGE BY AREA UNDER PACKAGE  
SCALE:20X

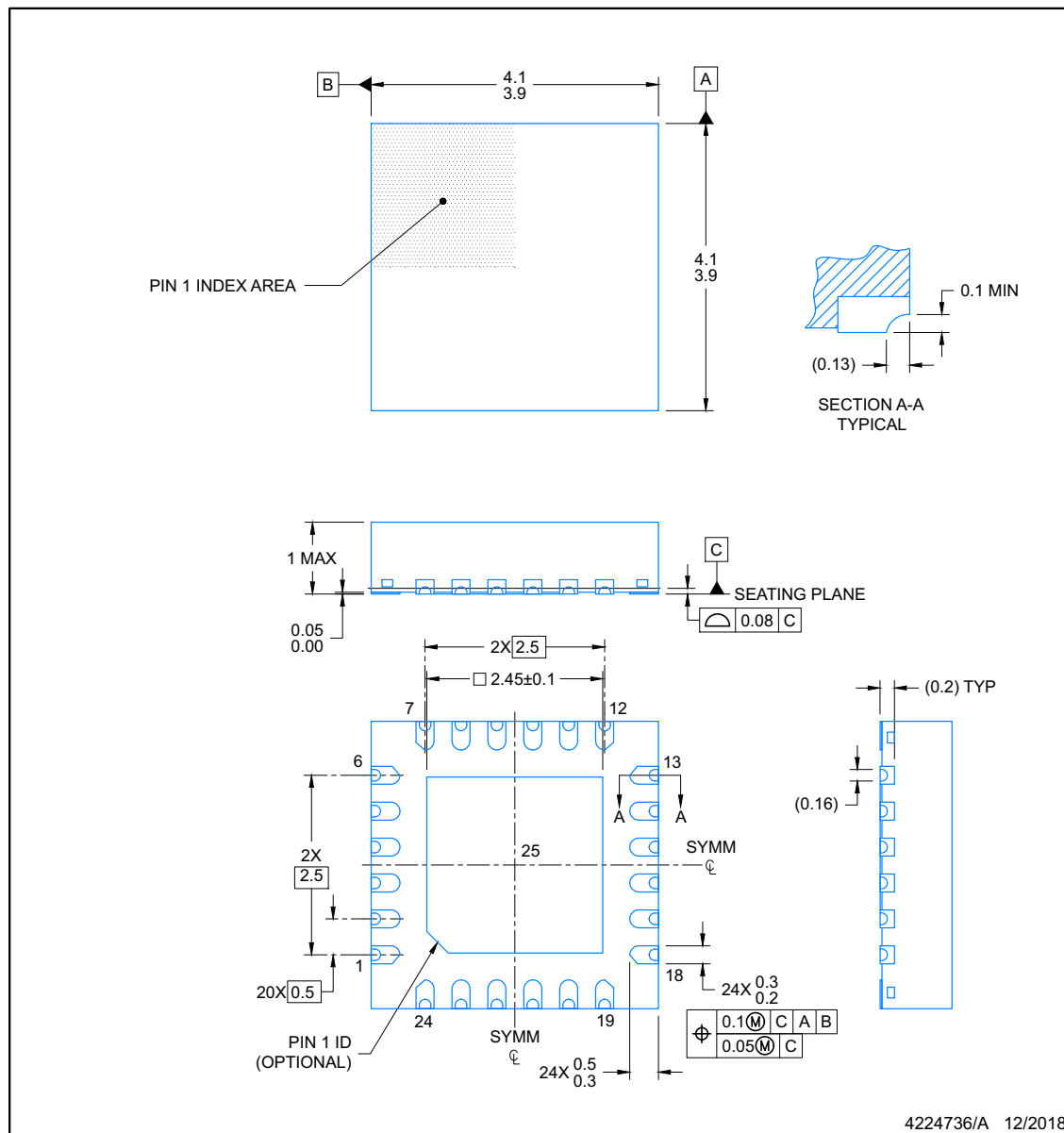
4219013/A 05/2017

NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

**RGE0024N****PACKAGE OUTLINE****VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK-NO LEAD

**NOTES:**

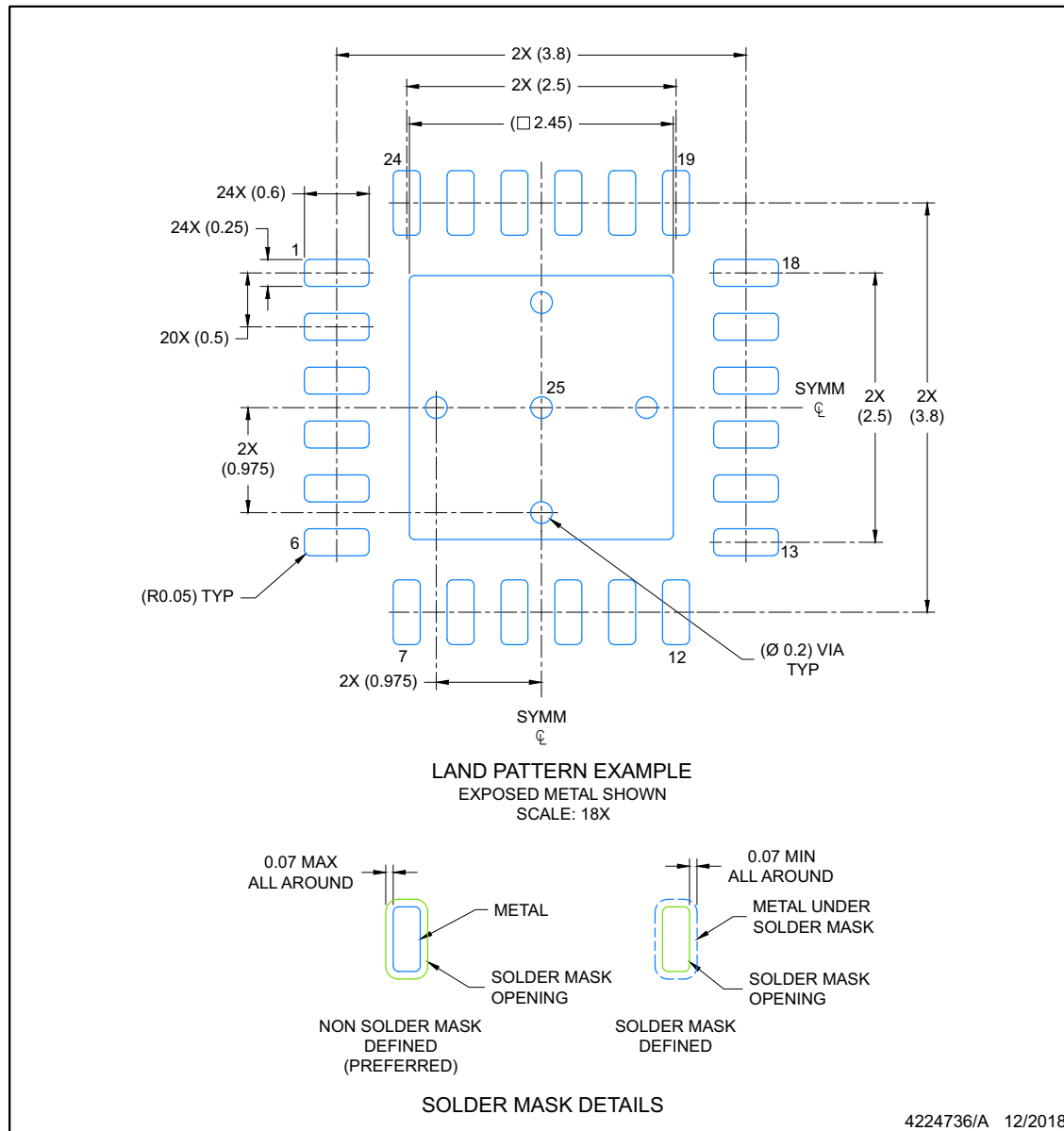
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for optimal thermal and mechanical performance.

## EXAMPLE BOARD LAYOUT

**RGE0024N**

**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK-NO LEAD



NOTES: (continued)

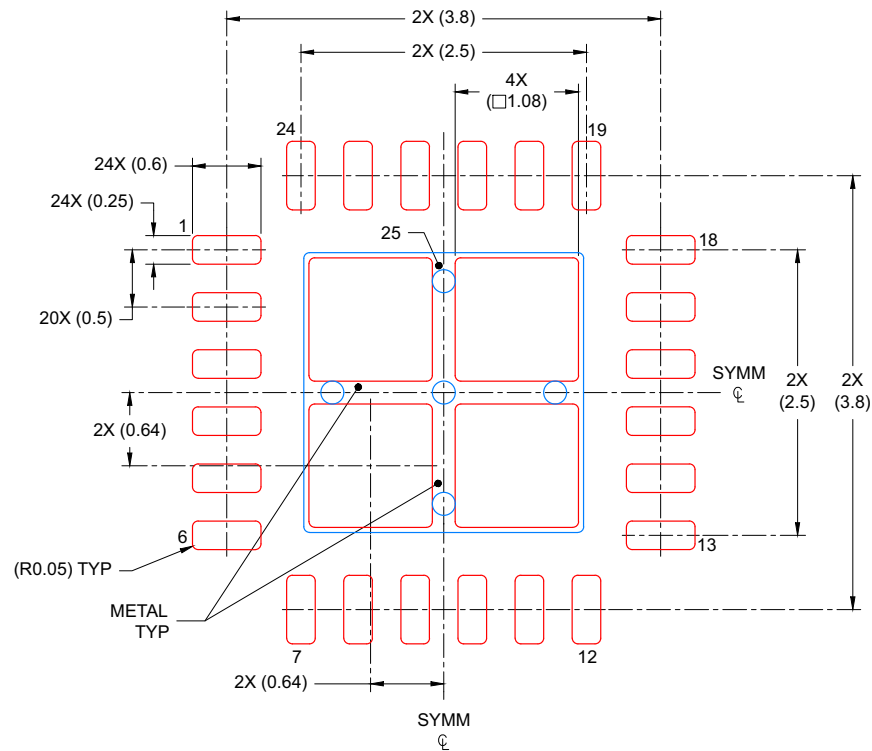
- This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 ([www.ti.com/lit/sluea271](http://www.ti.com/lit/sluea271)).
- Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

## EXAMPLE STENCIL DESIGN

**VQFN - 1 mm max height**

**RGE0024N**

PLASTIC QUAD FLATPACK-NO LEAD



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL

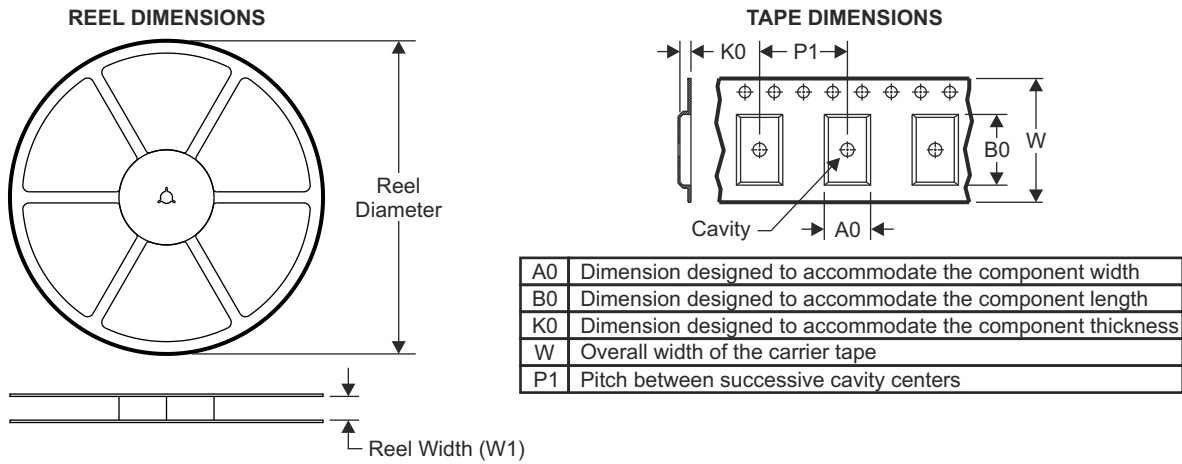
EXPOSED PAD  
78% PRINTED COVERAGE BY AREA  
SCALE: 18X

4224736/A 12/2018

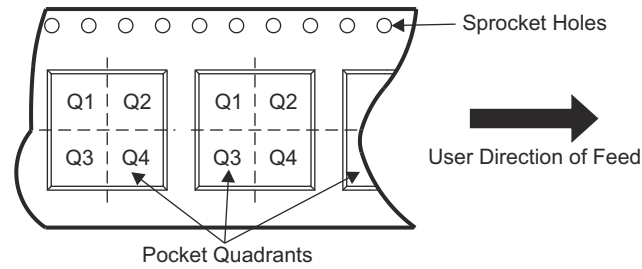
NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

## 11.2 Tape and Reel Information

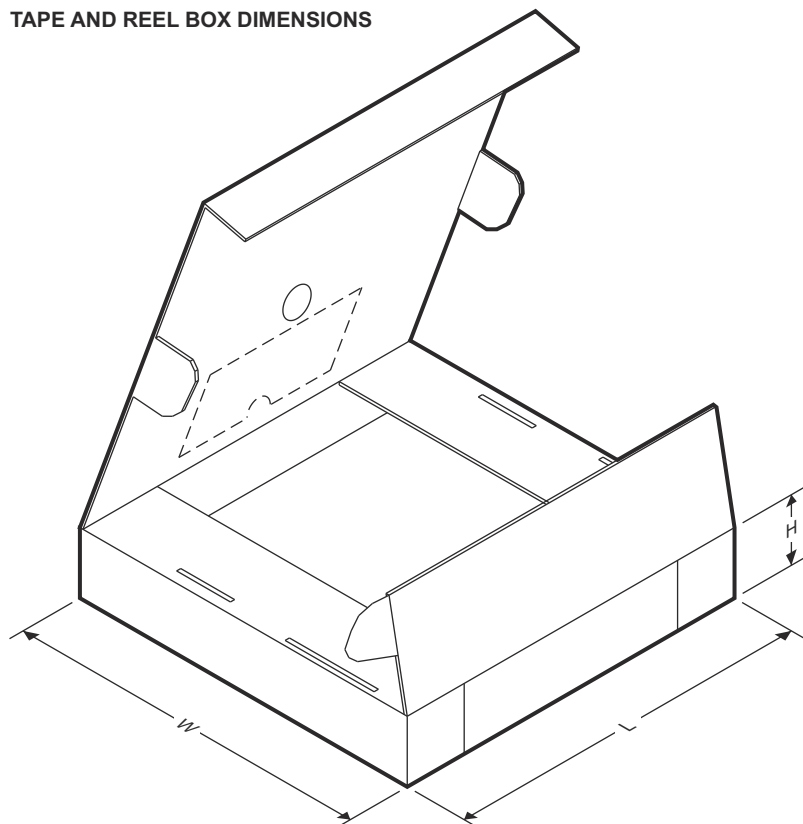


### QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



| Device           | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|------------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| TPS65131TRGERQ1  | VQFN         | RGE             | 24   | 3000 | 330.0              | 12.4               | 4.25    | 4.25    | 1.15    | 8.0     | 12.0   | Q2            |
| TPS65131WTRGERQ1 | VQFN         | RGE             | 24   | 3000 | 330.0              | 12.4               | 4.25    | 4.25    | 1.15    | 8.0     | 12.0   | Q2            |

## TAPE AND REEL BOX DIMENSIONS



| Device           | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|------------------|--------------|-----------------|------|------|-------------|------------|-------------|
| TPS65131TRGERQ1  | VQFN         | RGE             | 24   | 3000 | 356.0       | 356.0      | 35.0        |
| TPS65131WTRGERQ1 | VQFN         | RGE             | 24   | 3000 | 360.0       | 360.0      | 36.0        |



## PACKAGING INFORMATION

| Orderable part number            | Status<br>(1) | Material type<br>(2) | Package   Pins  | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|----------------------------------|---------------|----------------------|-----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| <a href="#">TPS65131TRGERQ1</a>  | Active        | Production           | VQFN (RGE)   24 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 105   | 2U65131<br>Q1       |
| TPS65131TRGERQ1.A                | Active        | Production           | VQFN (RGE)   24 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 105   | 2U65131<br>Q1       |
| <a href="#">TPS65131WTRGERQ1</a> | Active        | Production           | VQFN (RGE)   24 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 105   | 65131W<br>Q1        |
| TPS65131WTRGERQ1.A               | Active        | Production           | VQFN (RGE)   24 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 105   | 65131W<br>Q1        |

<sup>(1)</sup> **Status:** For more details on status, see our [product life cycle](#).

<sup>(2)</sup> **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

<sup>(3)</sup> **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

<sup>(4)</sup> **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

<sup>(5)</sup> **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

<sup>(6)</sup> **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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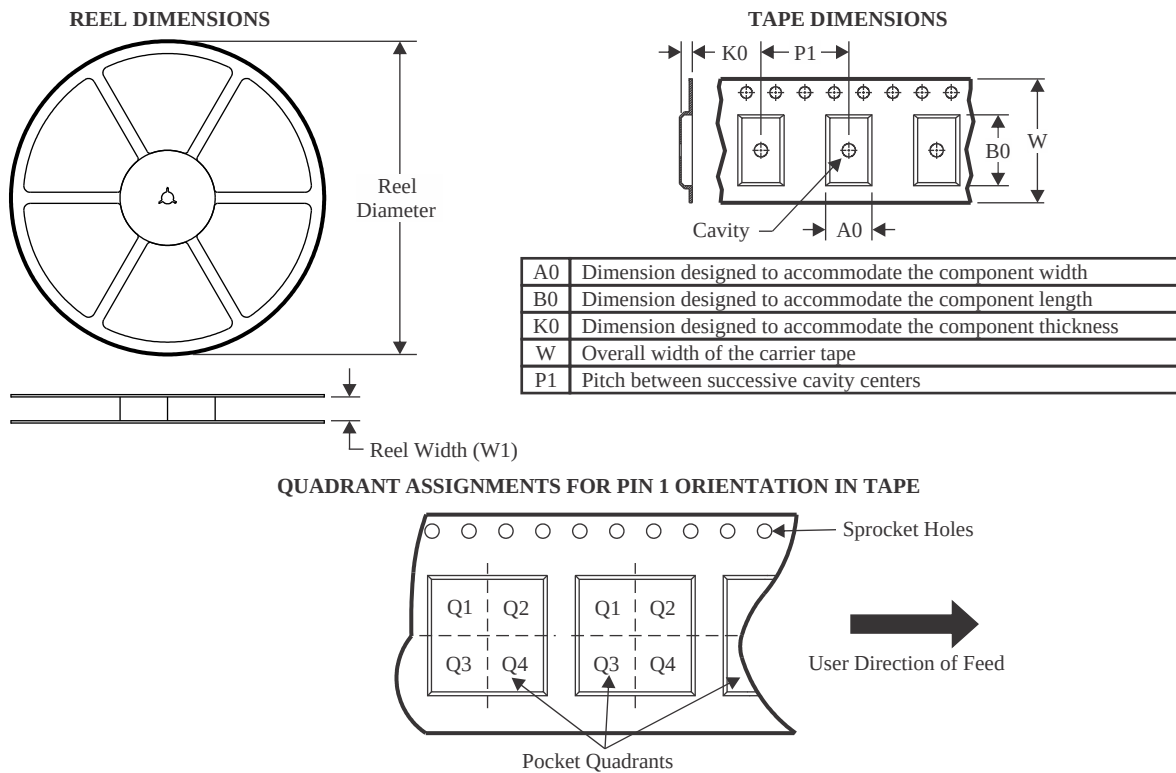
**OTHER QUALIFIED VERSIONS OF TPS65131-Q1 :**

- Catalog : [TPS65131](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product

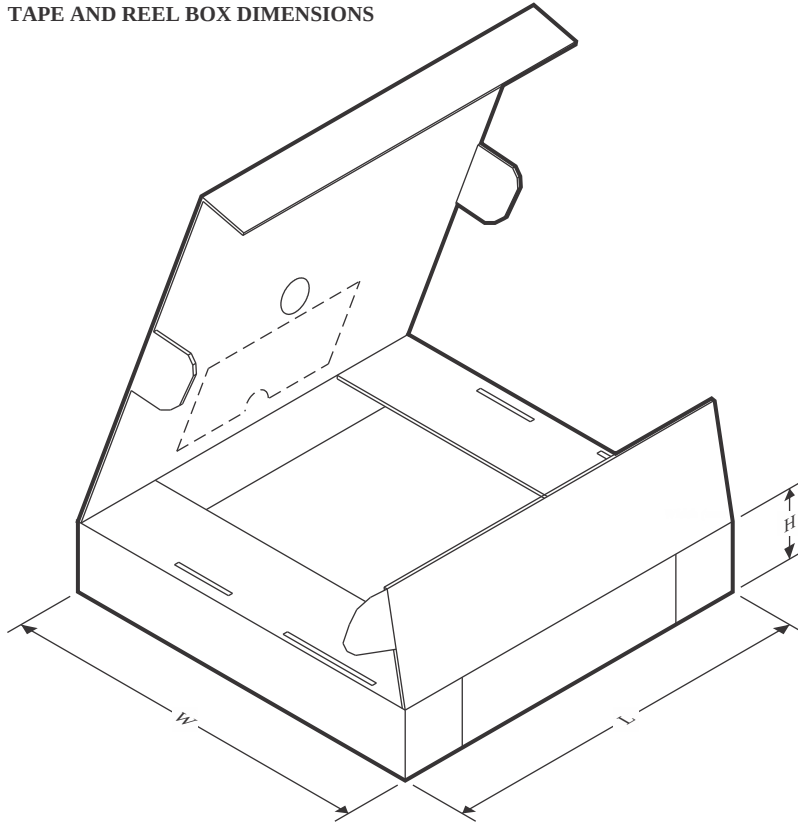
## TAPE AND REEL INFORMATION



\*All dimensions are nominal

| Device           | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|------------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| TPS65131TRGERQ1  | VQFN         | RGE             | 24   | 3000 | 330.0              | 12.4               | 4.25    | 4.25    | 1.15    | 8.0     | 12.0   | Q2            |
| TPS65131WTRGERQ1 | VQFN         | RGE             | 24   | 3000 | 330.0              | 12.4               | 4.25    | 4.25    | 1.15    | 8.0     | 12.0   | Q2            |

## TAPE AND REEL BOX DIMENSIONS



\*All dimensions are nominal

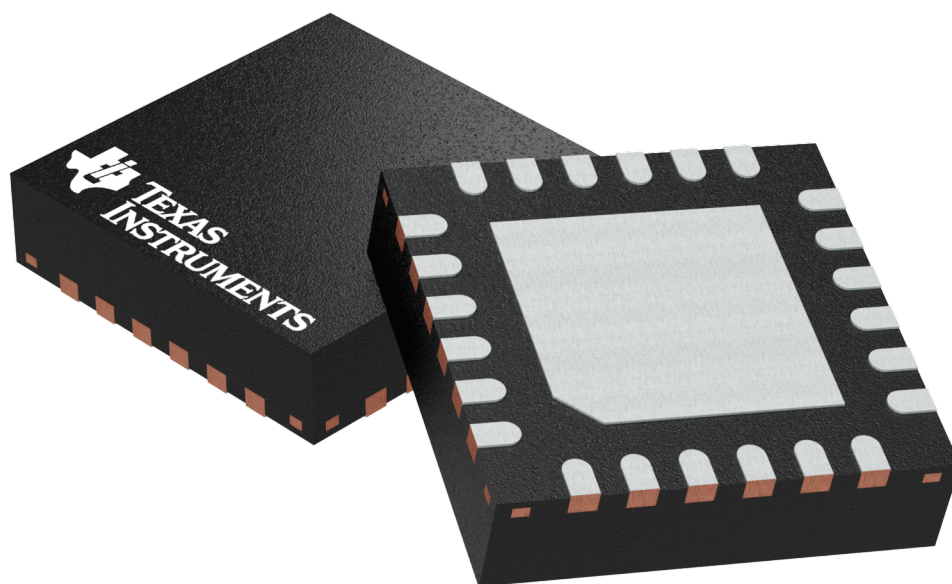
| Device           | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|------------------|--------------|-----------------|------|------|-------------|------------|-------------|
| TPS65131TRGERQ1  | VQFN         | RGE             | 24   | 3000 | 353.0       | 353.0      | 32.0        |
| TPS65131WTRGERQ1 | VQFN         | RGE             | 24   | 3000 | 360.0       | 360.0      | 36.0        |

**RGE 24**

**GENERIC PACKAGE VIEW**

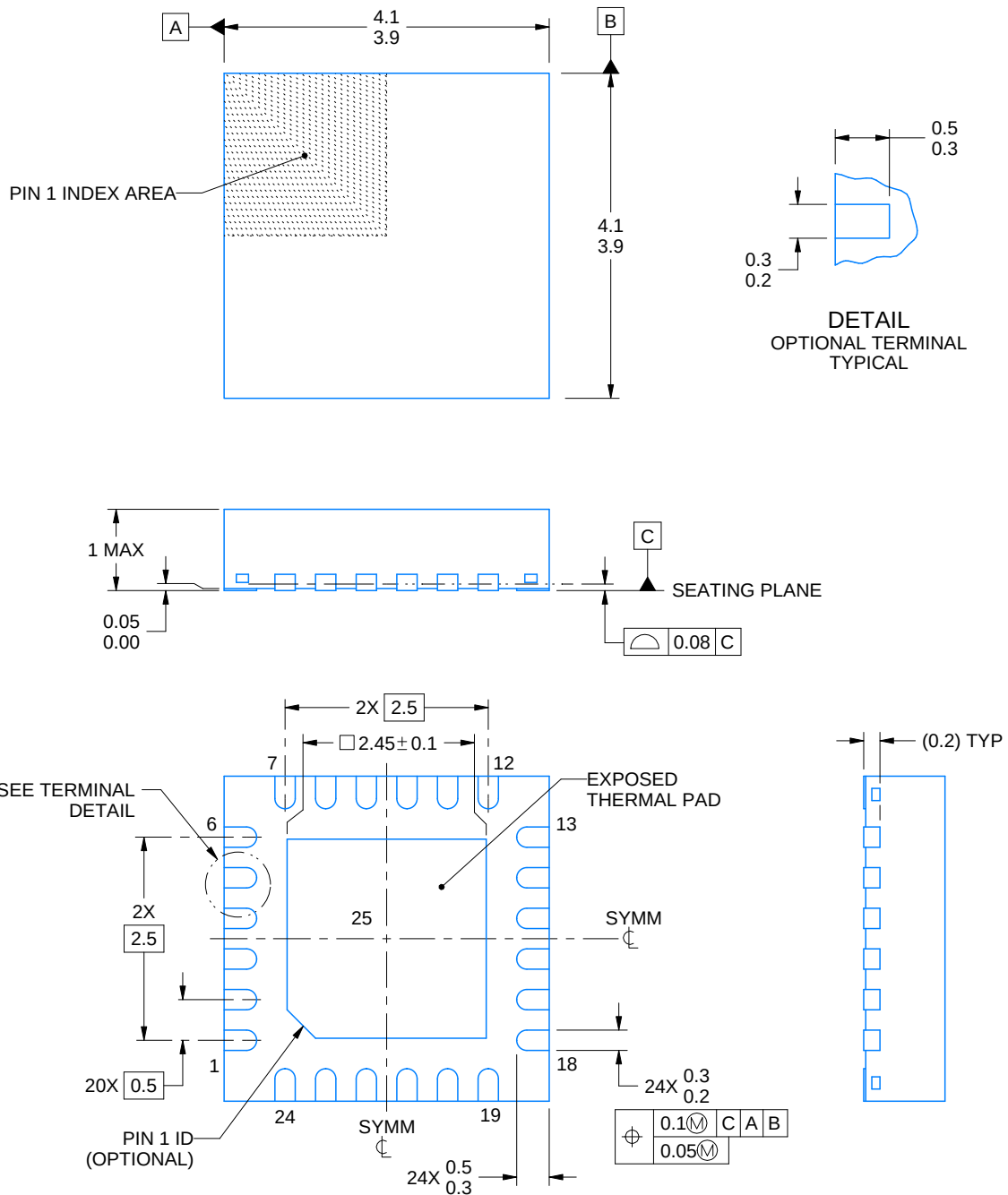
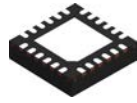
**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK - NO LEAD



Images above are just a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.

4204104/H



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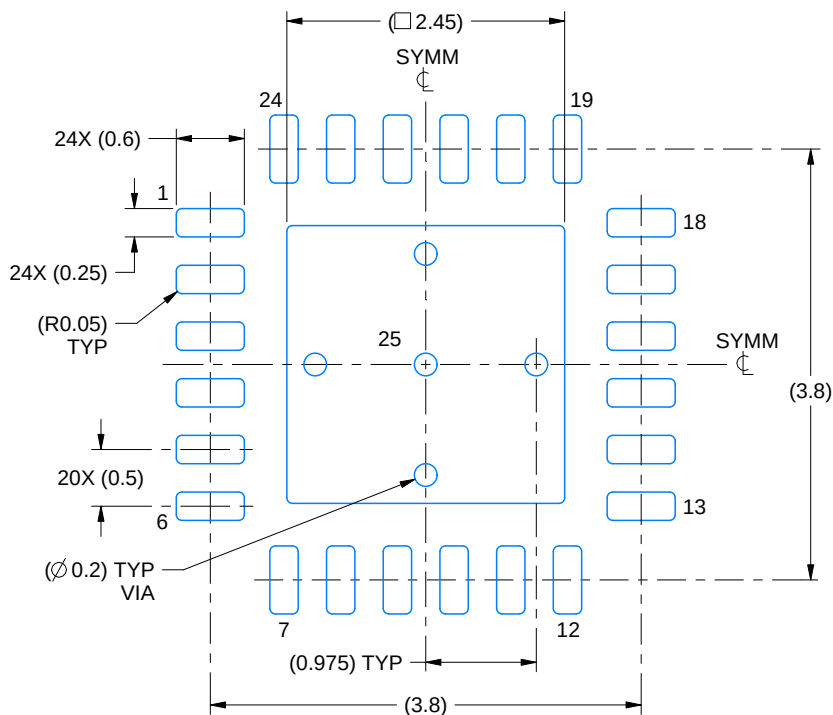
## NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for thermal and mechanical performance.

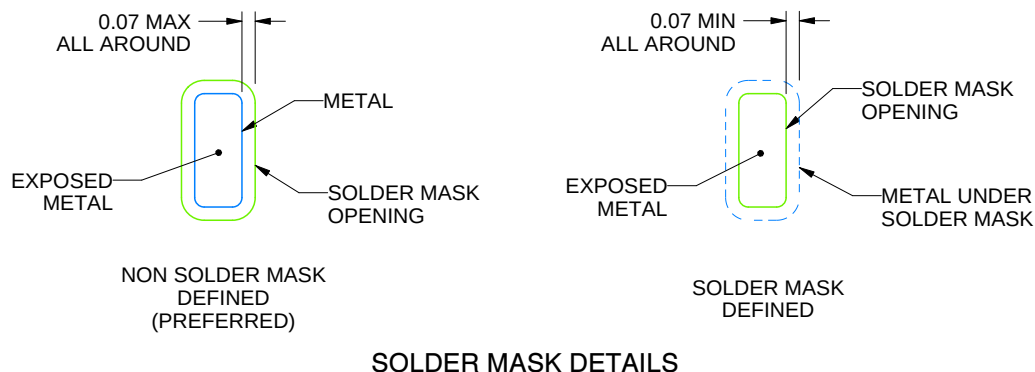
**RGE0024B**

**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK - NO LEAD



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:15X



4219013/A 05/2017

NOTES: (continued)

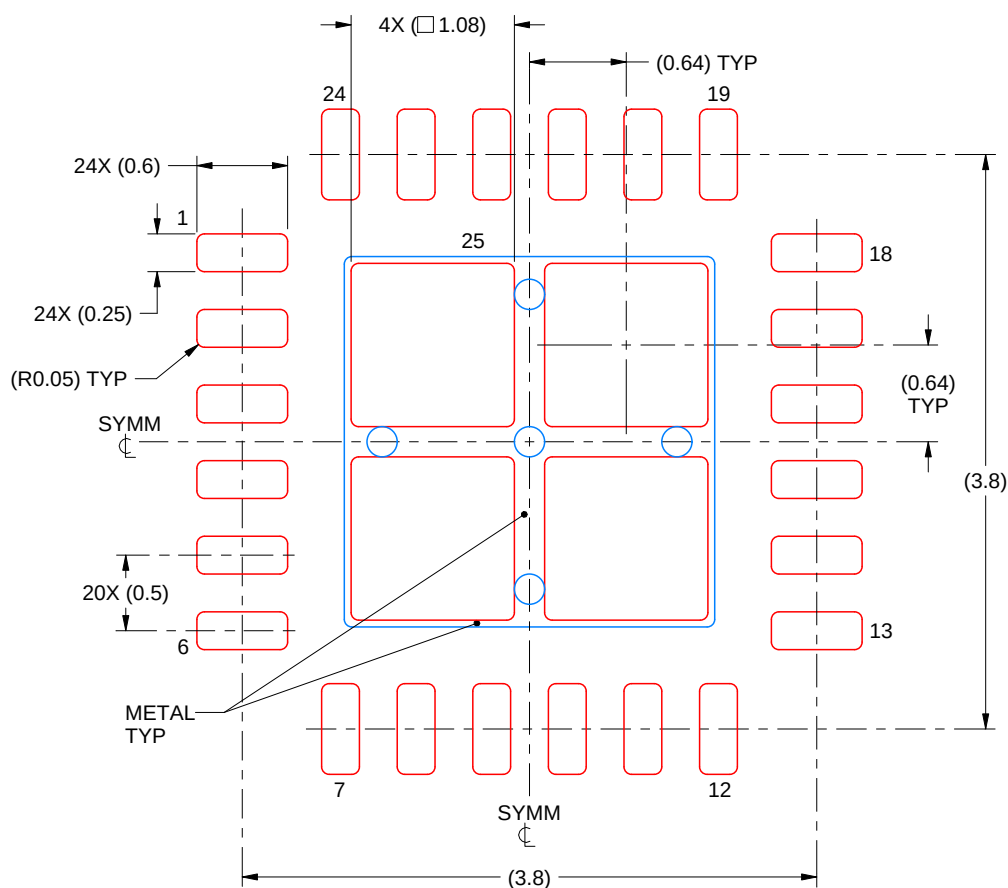
4. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 ([www.ti.com/lit/slua271](http://www.ti.com/lit/slua271)).
5. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

# EXAMPLE STENCIL DESIGN

RGE0024B

VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD



**SOLDER PASTE EXAMPLE**  
BASED ON 0.125 mm THICK STENCIL

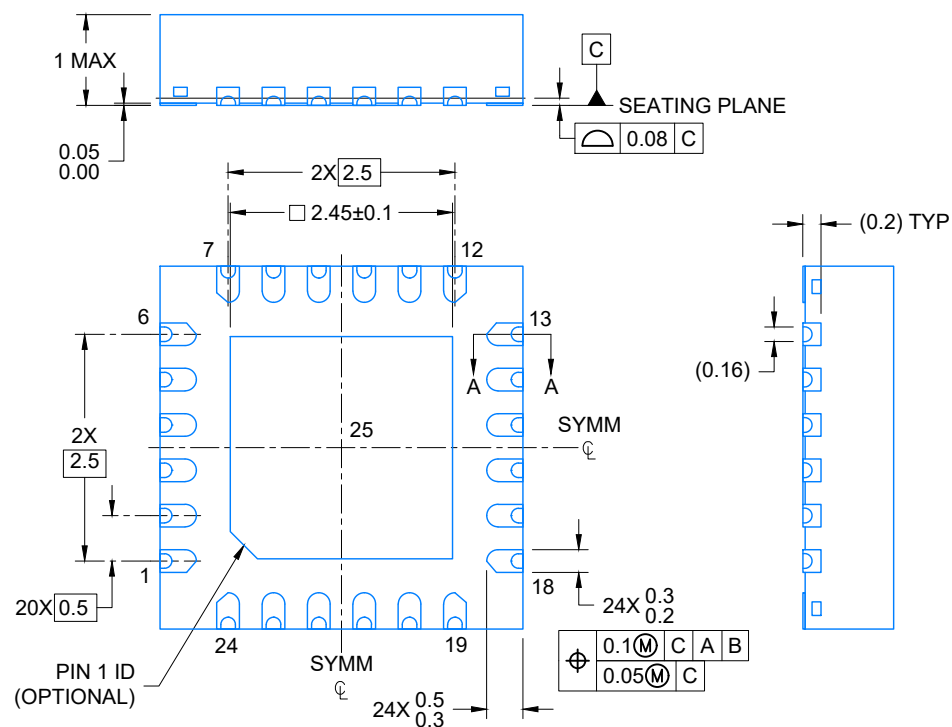
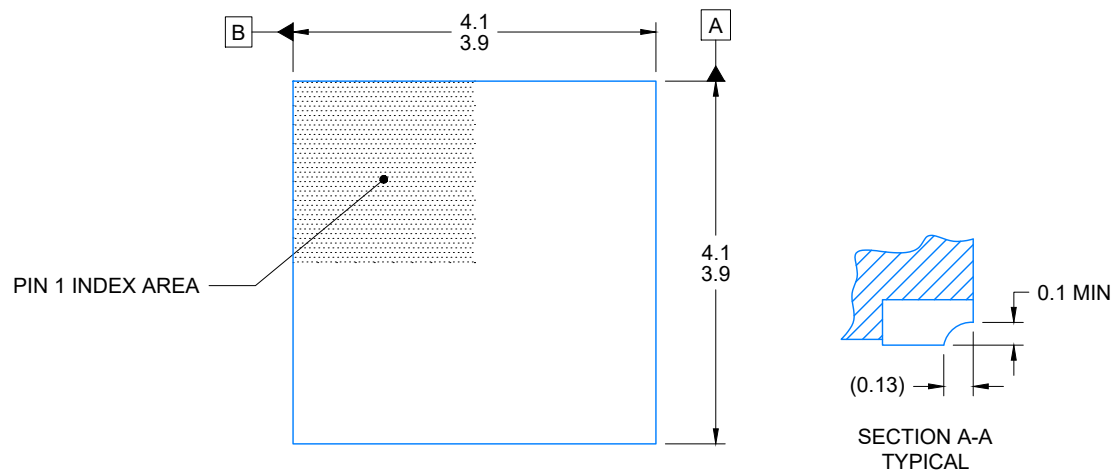
EXPOSED PAD 25  
78% PRINTED SOLDER COVERAGE BY AREA UNDER PACKAGE  
SCALE:20X

4219013/A 05/2017

NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.





4224736/A 12/2018

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for optimal thermal and mechanical performance.

## VQFN - 1 mm max height

24X (3.8)

2X (2.5)

(□ 2.45)

24

19

24X (0.6)

24X (0.25)

1

20X (0.5)

18

2X (0.975)

2X (2.5)

2X (3.8)

SYMM

25

13

6

(R0.05) TYP

7

12

2X (0.975)

(Ø 0.2) VIA TYP

SYMM

The diagram compares two PCB manufacturing methods:

- NON SOLDER MASK DEFINED (PREFERRED):** This method shows a rectangular opening in the solder mask. The opening is defined by a blue line, and the surrounding area is labeled "METAL". The solder mask opening is indicated by a green line. A dimension of "0.07 MAX ALL AROUND" is shown for the metal area.
- SOLDER MASK DEFINED:** This method shows a rectangular opening in the solder mask. The opening is defined by a green line, and the surrounding area is labeled "METAL UNDER SOLDER MASK". The solder mask opening is indicated by a blue line. A dimension of "0.07 MIN ALL AROUND" is shown for the metal area.

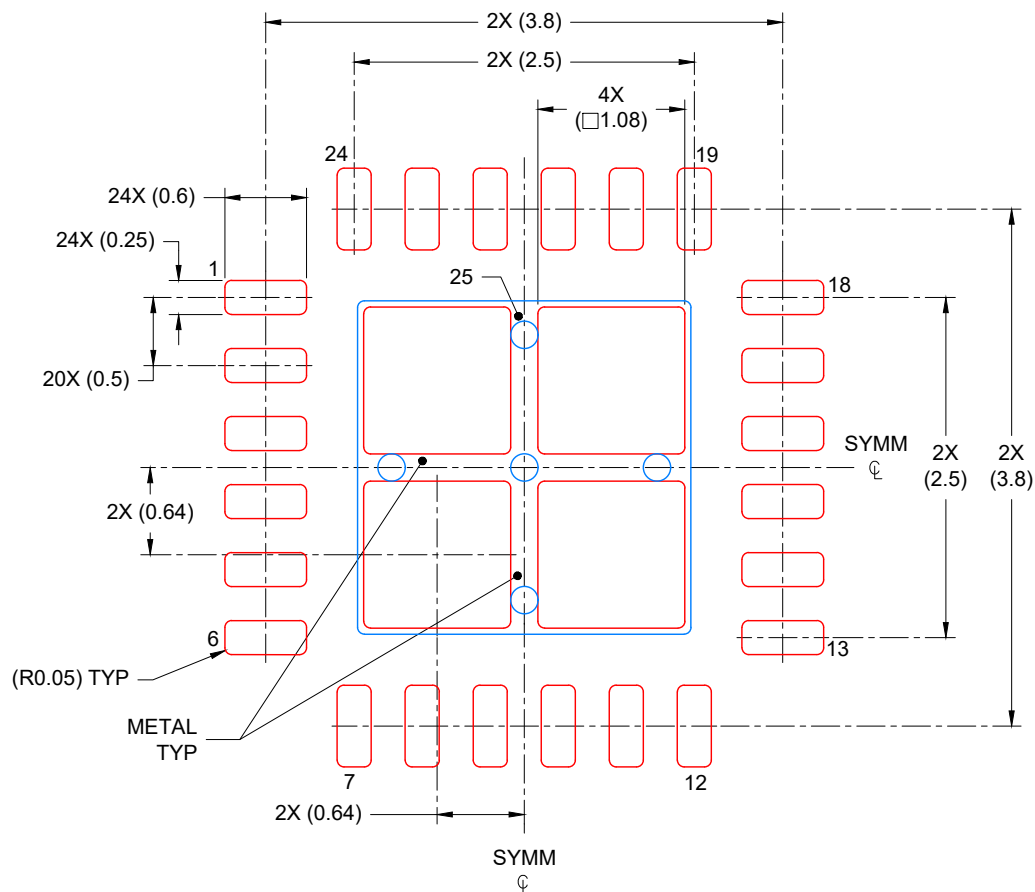
4224736/A 12/2018

4. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 ([www.ti.com/lit/slua271](http://www.ti.com/lit/slua271)).
5. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

**RGE0024N**

**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK-NO LEAD



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL

EXPOSED PAD  
78% PRINTED COVERAGE BY AREA  
SCALE: 18X

4224736/A 12/2018

NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

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